Chapter 4

Long-term effects of FTPM, CS and NR on music

Preface . . . 108
Abstract . . . 111
4.1 Introduction . . . 112
4.2 Methods . . . 119
4.2.1 Participants . . . 119
4.2.2 Stimuli . . . 120
4.2.3 Experimental Procedure . . . 121
4.2.4 Estimation of Frequency-to-Place Factors . . . 122
4.2.5 Data Analysis . . . 123
4.2.6 Statistical Analysis . . . 128
4.3 Results . . . 128
4.3.1 Analysis 0: Full-Sample Perceptual Trends . . . 129
4.3.2 Analysis 1: Interval Distortion and Chord Preference . . . 132
4.3.3 Analysis 2: Exploratory Electrode-Factor Associations . . . 139
4.4 Discussion . . . 145
4.4.1 Analysis 0: Full-Sample Perceptual Trends . . . 146
4.4.2 Analysis 1: Chord-Level Paired-Comparison . . . 147
4.4.3 Analysis 2: FIFs and Frequency-to-Place Factors . . . 149
4.4.4 Convergence of Analysis 1 and Analysis 2 . . . 150
4.4.5 Post-LCI Versus Pre-ECI Differences . . . 150
4.4.6 Limitations . . . 151
4.4.7 Clinical Implications . . . 152
4.5 Conclusion . . . 152
Postface . . . 153

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Preface

This chapter presents a preliminary version of the paper in preparation for submission to a peer-reviewed journal: **"Frequency importance function of CI users derived from a music-related task: relationship with frequency-to-place mismatch, neural responsiveness and current spread"**.

As described in Chapter 3 - Preface, the early pilots for both the vowel and music experiments followed a similar trajectory: a MUSHRA-based interface with realistic signals was initially adopted, and subsequently abandoned because, among others, the lack of control over which electrodes carried the perceptually relevant information made it impossible to draw meaningful conclusions with respect to the individual factors studied here (*i.e.*, FTPM, NR, and CS).

Critically, the limitations identified for speech stimuli in those earlier pilots were amplified in the music domain. Whereas speech pilots used a single signal type (*i.e.*, sentences from a female and male speaker), the music pilots employed stimuli drawn from different musical genres, with different levels of complexity (*e.g.*, different number of instruments) with and without lyrics, introducing additional variability in spectral content and temporal structure. Music signals are also inherently more complex than speech: they span a wider frequency range, contain richer harmonic structures, and engage overlapping perceptual dimensions (*e.g.*, pitch, timbre, rhythm, and harmony) that are difficult to disentangle.

As a result, the lack of electrode-level control that was already problematic for speech became untenable for music, reinforcing the need to move toward simpler, fully controlled stimuli.

The remainder of this preface summarises the pilot work that informed the final paradigm; the experiment reported in Sections 4.2–4.5 constitutes the main study. Note that the pilot numbering (Pilot 5b) continues the sequential numbering established in Chapter 3, where Pilots 1-5 covered the speech and early music development stages.

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Pilot 5b - Inverted chords with real piano recordings

Approach: This pilot was our first attempt to exploit the concept of chord inversion. A first-inversion chord contains the same pitch classes as the root-position chord, but with the root note raised by one octave so that it becomes the highest-pitched note in the triad - for example, the C major triad C-E-G in root position becomes E-G-C in first inversion. This displacement redistributes spectral energy to a higher cochlear region whilst preserving the same set of notes. In a paired comparison between a root-position chord and its inversion, a listener with uniform cochlear sensitivity should perceive both versions as equally pleasant (pleasantness is operationally defined in Section 4.2.3 as the participant subjective preference in the forced-choice paired comparison). However, if one version places energy in a cochlear region with distorted perception - for instance, due to FTPM or degraded ENI quality - it may be judged as less pleasant. By testing different combinations of inverted chords spanning the electrode array, this paradigm was intended to produce a perceptual profile reflecting the relative contribution of each cochlear region. Real piano recordings, bandpass-filtered to isolate target harmonics, were used as stimuli. It is worth noting that, because the stimuli were derived from real piano recordings, the resulting chords were not always traditional major or minor triads: the bandpass filtering sometimes isolated harmonics that did not correspond to conventional chord tones, and the harmonic spread within each chord depended on the richness of the original recording.

Evaluation:

- **Realistic signals:** *Partially satisfied* - the use of real piano recordings offered ecological validity, as chords embedded in natural timbral content are more representative of everyday music listening than synthetic tones. However, this approach faced the same challenge as the speech pilots: real chords selected according to each participant's electrode mapping did not always align well with individual frequency allocation tables, and consequently did not allow complete coverage of all electrode positions. Synthetic stimuli generated directly from each participant's map were also trialled in a subsequent iteration, and ultimately provided full electrode coverage, motivating the final design.
- **Clinical suitability:** *Satisfied* - the paired-comparison paradigm combined with a Bayesian model was well understood by participants and was considerably simpler and more enjoyable than a subjective rating of a single sound (as in the vowel task).
- **Electrode-level control:** *Partially satisfied* - coverage was achieved for most electrode positions, and the gap between covered regions never exceeded one electrode. However, coverage was not always complete for all participants, and the harmonic content delivered to each electrode was not fully controlled due to the variability in spectral spread across real piano recordings.

Decision: The paired-comparison paradigm and the chord inversion concept were retained, but real piano recordings were replaced by synthetic chords generated dynamically according to each participant's individual electrode mapping. This afforded full control over which electrodes were stimulated in each comparison, over the spectral content delivered to each electrode, and over the harmonic relationships between chord notes.

The final task presented in this chapter is the result of a progressive refinement - from a MUSHRA-based paradigm using realistic recordings, through a real-chord paired-comparison pilot, toward a fully synthetic and precisely controlled chord paradigm designed for electrode-level investigation.

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Frequency importance function of cochlear implant users derived from music-related task: relationship with frequency-to-place mismatch, neural responsiveness and current spread

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Abstract

Music quality and enjoyment are perceptual areas that CI users commonly report as priorities for improvement, even after years of device experience.

Whilst frequency-to-place mismatch (FTPM), current spread (CS), and neural responsiveness (NR) are suspected to contribute to these difficulties, their influence on music-specific perception has rarely been examined at the level of individual electrodes. This chapter presents an original chord paired-comparison paradigm designed to probe electrode-level sensitivity to spectral distortion during music listening.

Twenty-four CI users (fourteen late-implanted, Post-LCI; ten early-implanted, Pre-ECI) rated the pleasantness of chord pairs, each composed of a root-position chord and its first inversion, synthesised to probe specific pairs of electrodes.

Three analyses were performed. Analysis 0 characterised perceptual trends across the full sample: electrode-level frequency importance functions (FIFs) were derived for all participants using a Bayesian paired-comparison model, revealing a systematic apical-to-basal preference gradient that was equivalent across groups. Root-position preference, distortion sensitivity, and cue integration were also compared between Post-LCIs and Pre-ECIs. Analysis 1 examined the effect of FTPM-induced interval distortion on chord-level preference in the subset of participants with available FTPM estimates ($n = 7$). Analysis 2 related the electrode-level FIFs to individual FTPM, CS, and NR estimates in the subset of participants with FTPM estimates ($n = 7$) and PECAP data ($n = 5$).

Results showed a dominant preference for root-position chords across all participants, irrespective of implantation history, most plausibly explained by the acoustic structure of the stimuli rather than individual FTPM profiles. Post-LCI and Pre-ECI users differed in the nature of their distortion sensitivity: Post-LCI users showed inversion-conditional distortion avoidance consistent with a cue integration account, whilst Pre-ECI participants were heterogeneous, with one exhibiting inversion-unconditional avoidance and one showing no distortion sensitivity. In the exploratory factor analyses, CS was the most consistent electrode-level predictor of perceptual preference weight, whilst FTPM showed

a significant positive association in Post-LCI users that was partly confounded with electrode position. Post-LCI users showed significantly greater electrode-level preference differentiation than Pre-ECI users, and a significant FTPM × Group interaction confirmed that the two groups relate to their individual mismatch profiles in opposing directions. These findings provide the first evidence that FTPM-induced spectral distortion affects chord quality perception in CI users, and that this effect depends on both implantation history and the acoustic structure of the stimuli presented.

4.1 Introduction

Music quality and enjoyment are perceptual areas that CI users commonly report as priorities for improvement, even after years of device experience (Drennan et al., 2015). These two aspects of music are, however, fundamentally complex phenomena that cannot be reduced to acoustic fidelity alone. Musical enjoyment is shaped by cognitive and emotional processes - including expectation, cultural familiarity, and reward-circuit dynamics - rather than solely by the precision of spectral encoding (Huron, 2006; Zatorre and Salimpoor, 2013). The clearest evidence for this dissociation comes from congenital amusia, in which severe deficits in pitch discrimination coexist with broadly preserved emotional responses to music (Hyde and Peretz, 2004), demonstrating that spectral accuracy is neither necessary nor sufficient for musical pleasure in general. Consistent with this view, many CI users report continued enjoyment of music despite profound alterations to its spectral and timbral properties (Gfeller et al., 2008), and the relationship between measurable perceptual degradation and subjective musical pleasure remains poorly understood. It is important to note, therefore, that the present study does not address global musical pleasure - a question that implicates high-level cognitive and affective mechanisms largely independent of peripheral spectral coding. Instead, it targets a more constrained perceptual judgement: the relative quality of simultaneously presented chord versions, where quality is determined directly by how faithfully the CI conveys harmonic frequency relationships. This distinction is important, because spectral fidelity - unlike global pleasantness - is potentially addressable through targeted fitting strategies.

Unlike speech, for which accurate perception mainly relies on temporal envelope cues that CI devices transmit relatively well, music perception - specifically its harmonic and timbral dimensions - requires the accurate conveying of substantially more complex acoustic components - including fine spectral structure, pitch accuracy, and harmonic relationships - that current CI technology conveys poorly or may alter (McDermott, 2004; Limb and Rubinstein, 2012).

As a consequence, music quality - and in particular the perception of melody, harmony, and chord structure - is consistently rated among the most impaired aspects of auditory experience by adult CI users (Drennan et al., 2015). Moreover, it has been observed that such degradation is highly variable between individuals: some CI users report meaningful musical enjoyment, while others are profoundly affected (Gfeller et al., 2008), suggesting that individual factors play a significant role in shaping music perception outcomes.

Different categories of individual factors have been identified as contributing to this variability, including auditory history (*e.g.*, duration of deafness (Gfeller et al., 2008), residual acoustic hearing at the time of implantation (Gfeller et al., 2007), duration of CI use (Gfeller et al., 2008), and age at testing (Gfeller et al., 2008)), cognitive abilities (*e.g.*,

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working memory, processing speed, and attention (Gfeller et al., 2008)), and musical background and lifestyle (*e.g.*, pre-implant and post-implant musical training (Gfeller et al., 2008; Fuller et al., 2018), and post-implant musical habits (Gfeller et al., 2010)). These same factors have been shown to relate not only to speech perception but also to chord discrimination ability and music enjoyment, though not always in the same pattern as for speech (Lam et al., 2022).

An additional source of variability that cuts across all these factors is the timing of implantation relative to auditory experience. Post-LCIs developed auditory representations through normal acoustic hearing prior to their hearing loss, and must subsequently adapt those pre-existing representations to the constraints of the CI signal. Pre-ECIs, by contrast, never acquired such representations, their auditory representations having been shaped entirely under the CI modality. Although music listening questionnaires report no difference in music sound quality or appreciation between these groups (Moran et al., 2016), the dissociation described above - degraded performance alongside higher appreciation in Pre-ECIs (Bruns et al., 2016; Fuller et al., 2019) - suggests that the two groups relate to their CI signal in fundamentally different ways, possibly reflecting the absence of a spectral reference frame in users who have never heard acoustically.

Of particular interest here is a separate source of variability that, unlike the factors described above, offers genuine perspectives for improving music perception outcomes: the frequency-to-place factors. These factors were previously distinguished as FTPM and ENI quality[1]. They are of clinical relevance precisely because they are, at least in theory, addressable through individualised fitting strategies.

FTPM is possibly a key contributor to music degradation among these factors, and a major source of inter-individual variability. Its effects on music perception arise mainly through a shift of the entire frequency spectrum information along the cochlea, a shift which has been evaluated at approximately 1.5 octaves (towards high frequencies) on average and which is highly variable between CI users (Canfarotta et al., 2020a). Crucially, this shift is not uniform: FTPM appears to be non-linear along the electrode array, with greater mismatch typically present at the most apical and basal electrodes than at electrodes located in the middle of the array (Landsberger et al., 2015; Stakhovskaya et al., 2007a). This non-linearity is important because, whilst the auditory system appears capable of adapting to a relatively uniform spectral shift (Baskent and Shannon, 2004), an inconsistent shift along the array might impose qualitatively different demands, disrupting the harmonic organisation of sound rather than merely transposing it. Such non-linear distortion has two particularly detrimental effects for music: first, it may disrupt the harmonic relationships between notes, an aspect which is critical for melody and chord quality perception (a mechanism demonstrated in the study presented in Chapter 2); secondly, the distortion concentrated at the most apical electrodes is especially problematic for music, since crucial musical cues - including the fundamental frequency, melodic contour, lower harmonics, and timbral richness - are encoded by these apical electrodes, and it has been extensively shown that low-frequency information plays a substantial role in sound quality perception in CI users (Galvin et al., 2016; Caldwell et al., 2017a).

An important question is therefore whether, and for how long, FTPM continues to

[1]ENI encompasses CS (current spread) and NR (neural responsiveness). Detailed definitions are provided in the preceding chapters. In this chapter, we focus specifically on how these factors contribute to music degradation.

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affect music perception in experienced users. For speech comprehension (which constitutes to date the majority of interest in the research literature), the available evidence suggests that the perceptual consequences of FTPM diminish as users adapt over the first year of device use: Mertens et al. (2022) found that FTPM correlated with speech-in-noise performance at six months but that this effect was no longer significant at twelve months, consistent with the central adaptation reported in earlier work (Canfarotta et al., 2020a). The apparent capacity for adaptation to spectral shift is also supported at the level of pitch and channel discrimination, which Mathew et al. (2017) showed improves over the first six months of device use and correlates with speech perception ability. For music perception - specifically melody recognition, pitch discrimination, and chord quality - however, no such adaptation appears to take place. Longitudinal studies in experienced CI users demonstrate that tasks relying on these musical dimensions do not improve spontaneously over time (Gfeller et al., 2010; Sulibhavi et al., 2022).

Whether central adaptation similarly compensates for the perceptual consequences of FTPM for music, as it appears to for speech, therefore remains unknown. A likely explanation for this dissociation is that speech understanding and sound quality are not equivalent: CI users may achieve sufficient speech intelligibility through adaptation to a spectrally distorted signal, yet sound quality - which reflects the fidelity of the perceived frequency representation rather than the ability to extract meaningful sense - remains substantially degraded even in experienced users (Berg et al., 2025b; Caldwell et al., 2017a). Music perception is fundamentally dependent on this sound quality dimension: accurate pitch relationships, harmonic coherence, and timbral fidelity are all determined by the precision of spectral encoding, and are thus directly vulnerable to FTPM-induced distortion in ways that speech understanding is not. Even if there is no strong direct evidence that FTPM is a persistent contributor to music degradation, some recent findings suggest that reducing FTPM through CT-based tonotopic fitting acutely improves music perception outcomes in new CI users (Creff et al., 2024) - whether a similar effect persists in experienced users remains to be established.

In addition to the absence of strong evidence linking music perception to FTPM, existing studies relating FTPM to performance share a common limitation: they characterise FTPM as a single value - typically the mean across the array or related to a specific electrode range - and link this average measure to speech performance (Mertens et al., 2022; Canfarotta et al., 2020a; Venail et al., 2015a) or music scores (Creff et al., 2024; Jiam et al., 2019). Such an approach cannot specifically identify which electrodes are most responsible for music degradation. The electrode-level relationship between FTPM and music perception therefore remains uncharacterised - a gap in knowledge that is of direct clinical relevance, since targeted fitting interventions would require knowledge of which electrodes drive the degradation.

Beyond FTPM, the quality of the ENI (*i.e.*, the functional coupling between each electrode and the local neural population it targets) may represent another source of music perception variability that may interact with FTPM. As part of this study, we examine two aspects of ENI quality, namely CS and NR. CS broadens the spatial extent of neural excitation, reducing spectral selectivity and increasing inter-channel interaction, thereby degrading the place-coding resolution on which pitch and timbre perception depend (Friesen et al., 2001); greater CS at a given electrode may thus amplify the perceptual consequences of any FTPM at that location. NR - broadly reflecting the health and density of surviving spiral ganglion neurons - modulates the effective site of stimula-

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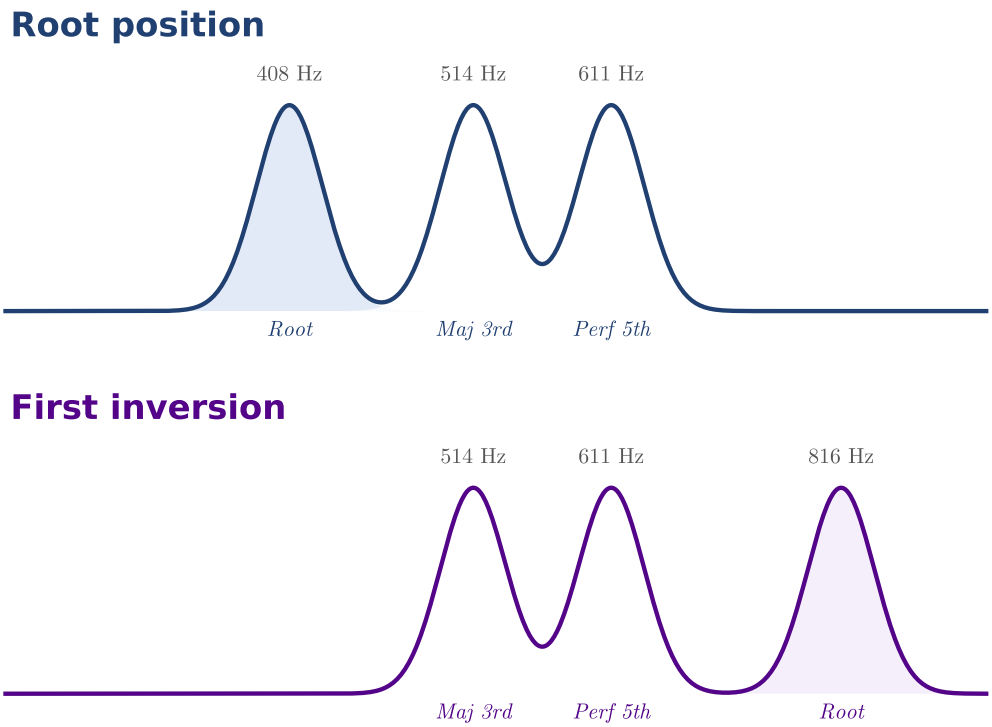


Figure 4.1: **Spectral structure of a root-position and first-inversion major chord.** Each panel shows the frequency components of a major triad built on a root of 408 Hz, tuned in 12-tone equal temperament. *Top:* root-position chord, comprising the root (408 Hz), major third (514 Hz), and perfect fifth (611 Hz). *Bottom:* first-inversion chord, in which the root note is displaced to the top of the chord (816 Hz), while the major third and perfect fifth remain unchanged. The two chords share two of three note frequencies; only the position of the root note differs. Comparing these two chords in a paired-comparison paradigm isolates the perceptual contribution of the electrodes stimulated by the displaced root note.

tion relative to the targeted cochlear location and the quality of information transmitted by a given electrode; where NR is degraded, neural activation may shift, broaden, or change in ways that compound the effects of both FTPM and CS (Garcia et al., 2021; Lambriks et al., 2023b). Despite the mechanistic plausibility of both these contributions, direct behavioural evidence linking ENI quality to music perception outcomes remains limited (Drennan and Rubinstein, 2008; Başkent et al., 2016), representing a further gap in knowledge in our understanding of what drives variability in music perception.

Together, these gaps motivate the development of an electrode-level perceptual tool capable of identifying which electrodes and associated cochlear regions are most affected by the frequency-to-place factors in terms of music perception - and that could ultimately be used to inform targeted remapping strategies. Achieving such electrode-level probing, however, would require a paradigm that affords precise control over which electrode carries the perceptually relevant information at any given moment - a requirement that realistic musical stimuli, with their associated temporal and spectral complexity, cannot satisfy.

To meet this requirement while preserving musical relevance, the present study uses chords as a stimulus unit. Chords occupy an intermediate position in the hierarchical organisation of Western music, situated between individual notes and complete musi-

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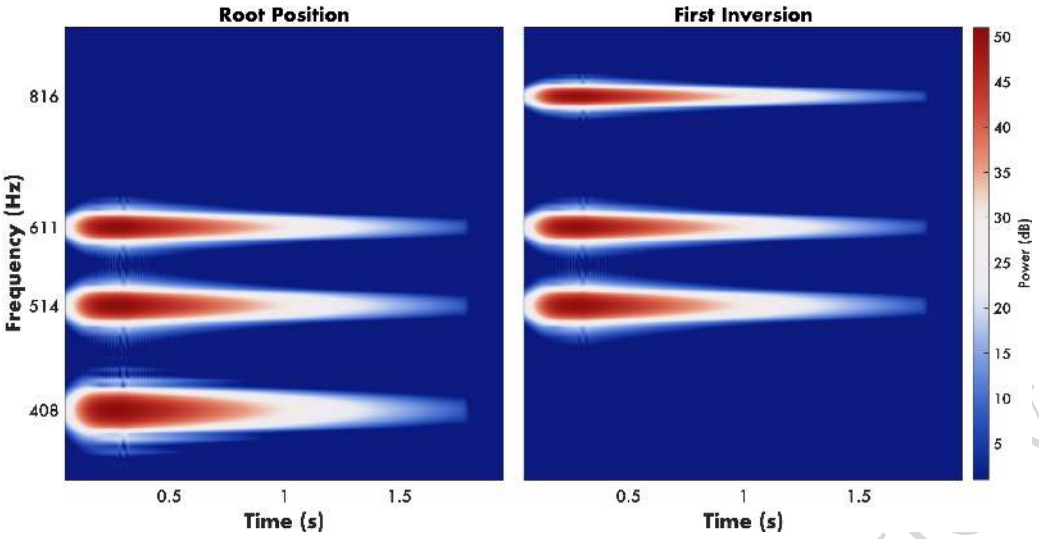


Figure 4.2: **Spectrogram representation of a root-position and first-inversion major chord.** Time-frequency representation of the same major triad as Figure 4.1 (root = 408 Hz, 12-tone equal temperament). *Left:* root-position chord (408, 514, 611 Hz). *Right:* first-inversion chord (514, 611, 816 Hz). The two chords share two frequency components (514 and 611 Hz); only the root note differs in octave position. The sigmoid attack and exponential decay envelope are visible in the temporal structure of each component. Colour scale indicates power in dB.

cal pieces. A chord can be defined as the simultaneous combination of notes at specific harmonic intervals - most commonly thirds in Western music - that together produce a perceptually coherent and pleasant harmonic unit (Plomp and Levelt, 1965). Throughout this study, stimuli consist exclusively of major triads, as major chords are among the most widely recognised harmonic structures and have been extensively used in the psychoacoustic literature on consonance and chord quality perception (Plomp and Levelt, 1965; Lahdelma and Eerola, 2022). The use of chords as a probe stimulus is further motivated by evidence that chord discrimination ability in CI users is systematically related to auditory history and device experience (Lam et al., 2022), suggesting that it captures clinically meaningful individual variability in the harmonic dimension of music perception.

Crucially, chord quality judgements are well suited to addressing the spectral fidelity question precisely because they are not reducible to global musical pleasure. Unlike musical enjoyment - which is shaped by expectation, cultural familiarity, and reward-circuit dynamics largely independent of peripheral spectral coding (Huron, 2006; Zatorre and Salimpoor, 2013) - the pleasantness of a chord version in a paired comparison task is determined directly by whether the harmonic frequency relationships between notes are faithfully conveyed to the auditory system. Chord quality therefore provides a tractable and ecologically grounded proxy for spectral fidelity, one that is sensitive to FTPM-induced distortion while remaining orthogonal to the broader and less tractable question of why humans find music enjoyable in the first place.

More specifically, we use the concept of first-inversion chords to probe music perception at the electrode level. In a first-inversion chord, the root note is displaced from the lowest to the highest position within the triad, effectively moving the root note up by one octave - for example, the C major triad C-E-G in root position becomes E-G-C in first inversion. This displacement redistributes the lowest spectral energy of the chord to a higher cochlear

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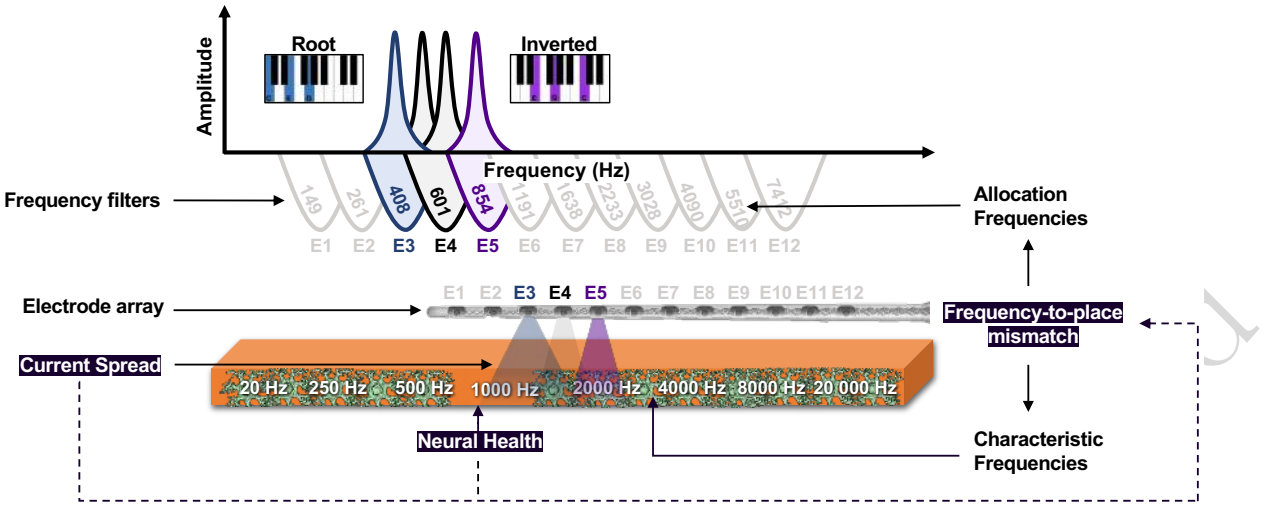


Figure 4.3: **Schematic illustration of the chord inversion paradigm and its potential relationship with the individual factors investigated.** An original chord and its inversion share the same notes, but differ in the octave position of the root note, thereby targeting different cochlear regions. The incoming signal is decomposed into frequency bands and assigned to electrodes E1–E12 according to the processor's frequency allocation table. Within the schematic representation of the cochlea (bottom), three individual factors determine the relationship between the allocated and the effectively perceived frequency. First, electrode array insertion depth sets the baseline FTPM by imposing a shift between characteristic frequencies and allocation frequencies. Second, current spread (coloured triangles) extends stimulation beyond the target electrode, recruiting neurons tuned to adjacent frequencies; its spatial extent is influenced by local neural health. Third, neural health varies along the cochlea; where spiral ganglion neurons are absent, the effective site of stimulation may shift to more distant surviving neural populations. The dotted lines between current spread and neural health indicate that these factors are suspected to modulate the primary FTPM effect rather than constituting independent sources of mismatch. In the example shown, the root chord's displaced note targets electrode E3, while the inverted chord's displaced note targets electrode E5, isolating the perceptual contribution of those two regions.

region whilst preserving the same three pitch classes. Figures 4.1 and 4.2 illustrate this principle for a major triad built on a root frequency of 408 Hz. The root-position chord comprises three simultaneous tones - the root (408 Hz), major third (514 Hz), and perfect fifth[2] (611 Hz). The first-inversion chord shares the major third and perfect fifth but displaces the root to the top of the chord (816 Hz), thereby redistributing spectral energy to a higher cochlear region whilst preserving the same three pitch classes.

For a listener with uniform sensitivity along the cochlea, both versions should be perceived as equally pleasant; however, if frequency-to-place factors place the root note of one version into a cochlear region where it disrupts the harmonic organisation or the proper conveying of the root note, that version may be judged as less pleasant than the other (see Figure 4.3 for a schematic overview of how these factors interact within the

[2]The term "perfect" denotes the interval class in standard music-theoretic nomenclature, not acoustic purity; in 12-tone equal temperament the perfect fifth is defined as 7 semitones ($2^{7/12} \approx 1.498$), which approximates but does not equal the just ratio of 3/2.

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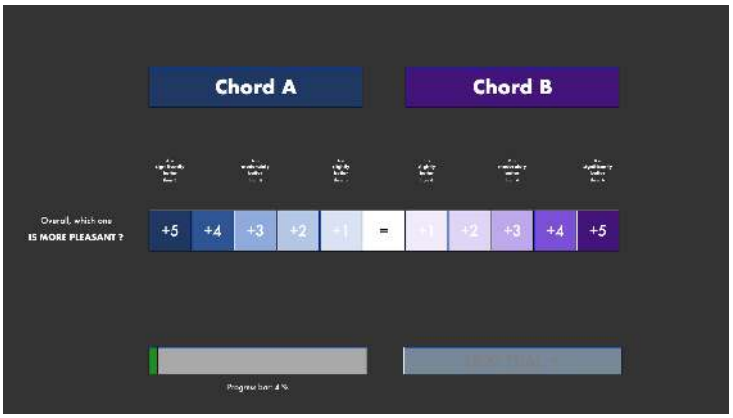


Figure 4.4: **Experimental interface for the chord paired comparison task.** On each trial, participants listened to two chords by pressing the Chord A (blue) and Chord B (purple) buttons at the top of the screen. After listening to both chords, participants indicated which they found more pleasant and the strength of their preference using an 11-point bipolar rating scale. The left side of the scale (blue) indicates a preference for Chord A and the right side (purple) a preference for Chord B, with labels ranging from "slightly better" (+1) through "moderately better" (+3) to "significantly better" (+5). The central "=" position indicates no preference. A progress bar at the bottom displayed the proportion of trials completed. Replay was permitted at any point during the trial.

paradigm used in this study).

On each trial, participants listened to both chord versions - the major chord triad and its first inversion - assigned to randomised positions across trials, with replay permitted at any point before responding. After listening to both chords, participants indicated their preference using an 11-point scale anchored at ±5 (*significantly better*), with intermediate ratings of ±3 (*moderately better*) and ±1 (*slightly better*), and a central position indicating no preference (=). The interface is illustrated in Figure 4.4.

Using this paradigm with multiple chord inversion comparisons probing all electrodes, carefully constructed according to each participant's individual electrode mapping, we address three complementary levels of analysis.

The first level of analysis (Analysis 0) characterises perceptual trends across the full sample of 24 participants (14 Post-LCI, 10 Pre-ECI), independently of individual frequency-to-place factor measurements. As part of this analysis, we derive electrode-level frequency importance functions (FIFs) for each participant using a Bayesian paired-comparison model, and examine whether the resulting profiles differ between Post-LCI and Pre-ECI groups in terms of which frequency regions are associated with higher or lower electrode-level sound quality. We also describe the overall pattern of chord preferences - including root-position preference, no-preference rate, and cue integration - and examine whether these perceptual tendencies differ between groups.

The second level of analysis (Analysis 1) examines, in the subset of participants with available FTPM estimates (5 Post-LCI and 2 Pre-ECI), whether chord preferences at the trial level can be explained by FTPM-induced disruption of harmonic organisation. Precisely, we compare at the trial level the degree of interval distortion introduced by FTPM between harmonics within each chord and relate this to the participant's choice (an illustration of the paradigm is proposed in Figure 4.5). We assume that, if FTPM-related

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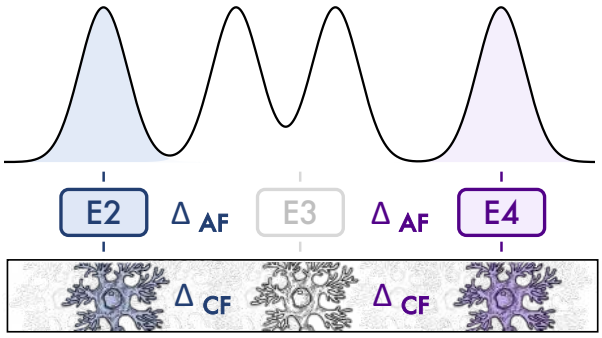


Figure 4.5: **Illustration of interval distortion introduced by FTPM and studied in Analysis 1.** A paired comparison opposes two chords sharing a common electrode; interval distortion arises when the cochlear-place interval Δ_{CF} differs from the allocated-frequency interval Δ_{AF}, *i.e.* when FTPM varies across electrodes.

distortion subsists, the chord chosen should be the one in which FTPM-related distortion is less pronounced. If confirmed, this would provide direct evidence that FTPM-related spectral distortion affects harmonic perception even after long-term experience with the device, suggesting that adaptation does not fully compensate the perceptual consequences of FTPM for music.

The third level of analysis (Analysis 2) relates the electrode-level FIFs derived in Analysis 0 to individual frequency-to-place factor estimates (FTPM, CS, and NR), in the subset of participants with FTPM estimates and PECAP data (see Table 3.1). This exploratory analysis examines whether electrodes associated with greater absolute FTPM magnitude, broader CS, or lower NR receive reduced perceptual weight in the FIF profiles.

Across all analyses, we hypothesise that FTPM-induced distortions carry a greater and more persistent perceptual cost for Post-LCIs, whose internal reference retains a memory of normal acoustic tonotopy, leading to more systematic down-weighting of spectrally distorted electrode regions in their FIF profiles and stronger preference effects in the paired comparison analysis. Pre-ECIs, having never developed auditory representations through normal acoustic hearing, may exhibit perceptual profiles shaped more directly by peripheral deficits (*i.e.*, CS and NR), with less sensitivity to FTPM-related distortions. Given the limited prior literature on this specific comparison, these are treated as directional hypotheses rather than established predictions.

4.2 Methods

4.2.1 Participants

14 Post-Post-LCI users (8 female; mean age = 45.4 years, SD = 13.2; mean duration of CI use = 5.6 years, SD = 2.1) and 10 Pre-Pre-ECI users (6 female; mean age = 25.4 years, SD = 3.4; mean duration of CI use = 20.9 years, SD = 1.0) participated in the study. CI participants were recruited from the Emmeline Centre at Addenbrooke's Hospital (Cambridge, UK). All participants were native speakers of British English. The study was approved by the NHS ethical committee (IRAS number: 335150), and all participants gave informed consent before the experiment.

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4.2.2 Stimuli

Stimuli consisted of synthetic chords in which the three composite tones were related by a major triad interval structure, tuned in 12-tone equal temperament (12-TET; frequency ratios of $1 : 2^{4/12} : 2^{7/12} \approx 1 : 1.260 : 1.498$). Unlike conventional musical chords built on standard pitches (*e.g.*, middle C, D, E), the fundamental frequencies of the chord tones were determined algorithmically from each participant's individual frequency allocation map, ensuring that they aligned with the centre frequencies of specific electrodes in the participant's array. The spectral structure and time-frequency representation of two example chords are illustrated in Figures 4.1 and 4.2, respectively.

Chord Generation

Chord stimuli were constructed algorithmically for each participant based on their clinical frequency allocation map.

Electrode Pairing. Electrodes were paired such that the centre frequencies of the two electrodes in each pair were approximately one octave apart (frequency ratio ≈ 2). More precisely, each pair consisted of two electrodes whose centre frequencies differed by a factor close to 2, so that the lower electrode of each pair served as the root of a root-position chord whilst the upper electrode served as the root of the corresponding first-inversion chord. The pairing algorithm iterated over a tolerance range from 5% to 30% (in 1% steps) until every active electrode appeared in at least one pair, with gaps between covered electrode regions never exceeding one electrode position. The final tolerance was participant-dependent, reflecting differences in frequency allocation across devices and maps. Because the number of active electrodes varied across participants and devices (12 for MED-EL, 16 for Advanced Bionics, 20–22 for Cochlear), the number of electrode pairs - and therefore the number of unique chord comparisons - differed between participants.

Chord Construction. For each electrode pair, two chords were generated: a root-position major triad whose root frequency was the centre frequency of the lower electrode, and a first-inversion major triad in which the root was displaced up by one octave to the centre frequency of the upper electrode. Note frequencies were computed in 12-TET. Given a root frequency f, the root-position chord comprised three simultaneous tones at:

$$f_{\text{root}}, \quad f_{\text{root}} \times 2^{4/12}, \quad f_{\text{root}} \times 2^{7/12} \tag{4.1}$$

corresponding to the root, major third (4 semitones), and perfect fifth (7 semitones). The first inversion retained the third and fifth at their original frequencies and relocated the root note up by one octave:

$$f_{\text{root}} \times 2^{4/12}, \quad f_{\text{root}} \times 2^{7/12}, \quad f_{\text{root}} \times 2 \tag{4.2}$$

The two chords in each pair thus shared two of three note frequencies (the major third and perfect fifth), differing only in the octave position of the root note. Critically, this means that each chord version activated a different pair of electrodes: the root-position chord stimulated the lower electrode (carrying the root) and the electrode assigned to the

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third and fifth, whilst the first-inversion chord stimulated the upper electrode (carrying the displaced root) and the same shared electrode.

Note-to-Electrode Mapping and Collapse

An important consequence of the coarse electrode spacing in CIs is that the three notes of a major chord do not resolve onto three separate electrodes. Because the frequency separation between adjacent chord notes (a major third $\approx$ 4 semitones, a perfect fifth $\approx$ 7 semitones) is smaller than the bandwidth assigned to most electrodes, two of the three notes consistently mapped to the same electrode. For root-position chords, the root occupied one electrode while the third and fifth collapsed onto the adjacent electrode; for first-inversion chords, the third and fifth shared one electrode while the octave-displaced root occupied another. Each chord was therefore effectively rendered as a two-electrode stimulus, with the two active electrodes in each pair determining which cochlear regions were compared across a given trial.

4.2.3 Experimental Procedure

The experiment used a two-alternative forced-choice paired comparison paradigm. On each trial, participants heard two chords - one in root position and one in first inversion - built around different electrode positions along the array. Each electrode pair was presented five times, yielding a total of $5 \times n_{\text{pairs}}$ trials per participant. Short breaks were offered every eight trials. Replay was allowed at any point during a trial.

Each chord was synthesised as the sum of three pure tones at the specified note frequencies, each shaped by a piano-like amplitude envelope comprising a sigmoid attack (300 ms) and an exponential decay (1.5 s), and normalised to prevent clipping. Chords were 2 s long at a sampling rate of 48 kHz. On each trial, the two chords were presented successively, separated by an inter-stimulus interval of 500 ms, and a visual indicator on screen showed which chord was playing at any given moment.Chords were presented monaurally at 65 dB SPL, via a MacBook Pro, an RME Fireface UCX II sound card, and a Sennheiser HD 600 headset positioned over the clinical processor of each CI user.

Response Format

After listening to both chords, participants indicated which chord they found more pleasant and how strongly they preferred it, using a single bipolar rating scale (Figure 4.4). The scale ranged from +5 (*Chord A is significantly better*) through = (*no preference*) to +5 (*Chord B is significantly better*), with an intermediate anchor at ± 3 (*moderately better*), yielding an 11-point response range. This format captures both the direction and the strength of preference in a single response.

The assignment of root-position and inverted chords to the two listening positions was randomised independently on each trial. Trial order was fully randomised within each session.

For the purpose of the analyses described below, the bipolar rating was used differently at each level of analysis, in alignment with the corresponding hypotheses. For Analysis 0,

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the full signed grade was used directly to characterise perceptual trends across the sample, and the full graded response was used as input to the Bayesian paired-comparison model (Section 4.2.5) to derive electrode-level FIFs. For Analysis 1, the rating was decomposed into a binary preference (direction of the response) and a preference strength (absolute value of the rating, 1–5); trials on which participants selected the "=" position were excluded from the binary sensitivity check but retained in the graded models (coded as a signed grade of zero), as the zero value carries information - it reflects the absence of preference - and discarding these trials would systematically reduce the effective sample size for participants with high no-preference rates. For Analysis 2, the FIF profiles derived in Analysis 0 were related to individual factor estimates.

Trial and Session Structure

Each electrode pair was tested five times per session, with the assignment of chords to the two listening positions randomised independently on each repetition. The total number of trials per session was $5 \times n_{\text{pairs}}$, where n_{pairs} depended on the number of active electrodes in the participant's map. Some participants completed two sessions; their data were combined for analysis.

4.2.4 Estimation of Frequency-to-Place Factors

Three electrode-level individual factors were estimated for each CI participant: FTPM, CS, and NR. The estimation procedures are described in detail in Chapter 3 (Section 3.2.4); only a brief summary is provided here.

FTPM

FTPM was estimated from post-operative X-ray images using a semi-automatic algorithm combining electrode localisation, cochlear spiral fitting, and the Stakhovskaya frequency-position function (Stakhovskaya et al., 2007b). For each electrode, the mismatch was expressed in octaves as:

$$\text{FTPM}_e = \log_2 \left(\frac{\text{CF}_{\text{place},\,e}}{\text{CF}_{\text{assigned},\,e}} \right) \tag{4.3}$$

where $\text{CF}_{\text{place},\,e}$ is the characteristic frequency at the cochlear location of electrode e (estimated from post-operative imaging) and $\text{CF}_{\text{assigned},\,e}$ is the centre frequency of the allocation band assigned to that electrode in the clinical map. Negative values indicate a basal shift (*i.e.*, the electrode stimulates a cochlear region tuned to a higher frequency than the frequency allocated to it).

CS and NR

CS and NR were estimated using the PECAP method (Garcia et al., 2021), as described in Sections 3.2.4–3.2.4.

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4.2.5 Data Analysis

The data were analysed at three complementary levels, corresponding to the sub-goals described in the Introduction. Analysis 0 characterised perceptual trends across the full sample of 24 participants, independently of frequency-to-place factor measurements, and derived electrode-level FIFs from the paired-comparison responses. Analysis 1 examined FTPM-induced harmonic distortion effects on chord preference at the trial level, restricted to the subset of participants with available FTPM data. Analysis 2 related the electrode-level FIFs derived in Analysis 0 to individual factor estimates (FTPM, CS, NR) and listener group, in the subset of participants with available factor data (see Table 3.1).

Analysis 0 - Full-Sample Perceptual Trends and Frequency Importance Function

Analysis 0 was conducted on all 24 participants (14 Post-LCI, 10 Pre-ECI) with chord preference data, regardless of whether individual frequency-to-place factor estimates were available. It addressed four descriptive questions: (i) whether certain frequency regions were systematically associated with higher or lower electrode-level sound quality, and whether the resulting FIF profiles differed between Post-LCI and Pre-ECI groups; (ii) whether participants showed a systematic preference for root-position chords over their inversions; (iii) whether the rate of no-preference responses and the strength of preference grades differed between groups; and (iv) whether the root-preference magnitude differed between Post-LCI and Pre-ECI.

Electrode-Level Frequency Importance Functions. For each participant, paired comparison responses were analysed using `PairedCompCalc` (Leijon et al., 2019), which estimates latent quality parameters on an interval scale from ordinal graded judgements via maximum *a posteriori* (MAP) estimation with a Bayesian prior. The full 11-point bipolar rating was used as input; tied responses ("=", Grade = 0) were included and handled natively by the model, contributing a symmetric likelihood that provides no discriminating information between the two electrodes. For each participant, the model estimated a latent quality parameter π_e for each electrode e:

$$P(\text{prefer } i \text{ over } j) = \frac{\pi_i}{\pi_i + \pi_j} \tag{4.4}$$

The resulting $\log(\pi_e)$ values were normalised to yield preference scores, with higher values indicating electrodes more consistently and strongly preferred. The most apical electrode served as the Bayesian reference and was fixed at zero by construction; all preference scores were therefore expressed relative to this reference electrode.

To compare FIF profiles across participants with different devices and electrode counts, electrode positions were normalised to a common axis (0 = most apical, 1 = most basal) within each participant. LMMs with a random slope for normalised electrode position were fitted to test for a systematic preference gradient along the array and for group differences in FIF shape:

$$\text{Preference} \sim \text{ElecPos_norm} \times \text{Group} + (1 + \text{ElecPos_norm} \,|\, \text{Participant}) \tag{4.5}$$

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A random slope was included because the Bayesian reference electrode anchors every participant's profile at zero, collapsing between-participant intercept variance to near zero; the between-participant variability is therefore in the gradient of the profile rather than its overall level.

Root-Position Preference. For each participant, a root-preference rate was computed as the proportion of decided trials (Grade $\neq 0$) in which the root-position chord was chosen, correcting for the random assignment of root and inversion to the two listening positions. A binomial test against chance (50%) was applied per participant. At the group level, root-position preference was modelled using LMMs with signed grade as the dependent variable and S1_IsRoot as the fixed effect, where S1_IsRoot is a binary indicator for whether Sound 1 was in root position (1) or first inversion (0). Models were fitted separately for each group and with a group-by-inversion interaction term in the pooled model:

$$\text{SignedGradeS1} \sim \text{S1_IsRoot} \times \text{Group} + (1 \,|\, \text{Participant}) \tag{4.6}$$

No-Preference Rate and Grade Strength. The proportion of trials with Grade = 0 was computed per participant as the no-preference rate. The mean absolute grade on decided trials was computed as a measure of preference strength. Both metrics were compared between groups using LMMs:

$$\text{NoPrefNum} \sim \text{Group} + (1 \,|\, \text{Participant}) \tag{4.7}$$

$$\text{AbsGrade} \sim \text{Group} + (1 \,|\, \text{Participant}) \tag{4.8}$$

where `NoPrefNum` is a binary indicator (1 = no preference, 0 = decided) and `AbsGrade` is the absolute value of the grade on decided trials. Group effects were extracted from the `GroupCat_ECI` coefficient in each model, providing a direct LMM-based test of Post-LCI versus Pre-ECI differences.

Analysis 1 - Chord-Level Interval Distortion

Analysis 1 addressed whether experienced CI users prefer chord versions whose harmonic structure is less distorted by their individual FTPM profile. It was restricted to the seven participants with available FTPM data (five Post-LCI, two Pre-ECI) and operated at the level of individual paired comparison trials. Graded LMMs predicting the signed preference rating were the primary analysis; a binary GLLM served as a sensitivity check. The rationale for using graded models as primary is twofold: they retain all trials including no-preference responses (coded as 0), maximising statistical power; and they capture preference strength, not just direction, which provides more information about the magnitude of distortion sensitivity.

Electrode Mapping and Note Collapse. Each chord was mapped to two electrodes following the collapse described in Section 4.2.2.

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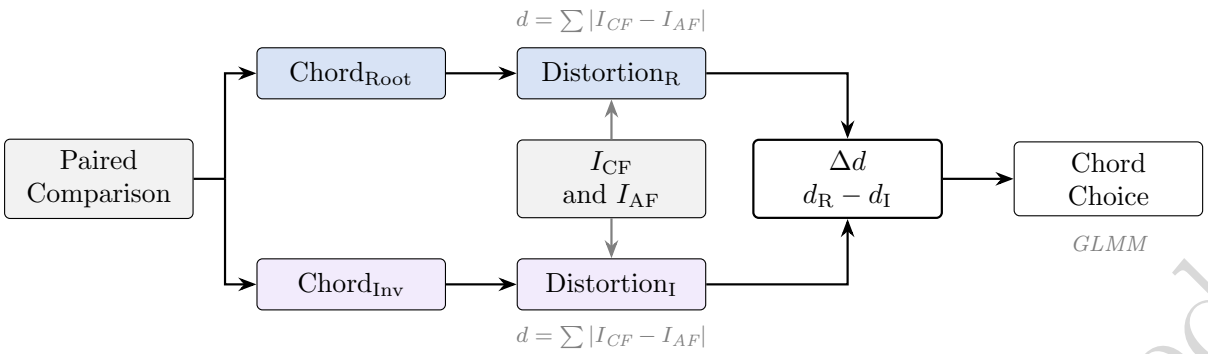


Figure 4.6: **Analysis pipeline for Analysis 1** Each paired comparison trial presents two chords. For each chord, the intended interval I_{AF} (based on allocated frequencies) is compared to the physical interval I_{CF} (based on cochlear place frequencies) to compute interval distortion ($d = \sum |I_{\text{CF}} - I_{\text{AF}}|$). Distortion reduces algebraically to the FTPM gradient across the two chord electrodes ($I_{\text{CF}} - I_{\text{AF}} = \text{FTPM}_j - \text{FTPM}_i$), such that a uniform mismatch across electrodes leaves the interval structure intact.

Interval Distortion Computation. For each chord, interval distortion was computed as:

$$d = \sum |I_{\text{CF}} - I_{\text{AF}}| \tag{4.9}$$

which reduces algebraically to the differential mismatch between the two chord electrodes:

$$I_{\text{CF}} - I_{\text{AF}} = \text{FTPM}_j - \text{FTPM}_i \tag{4.10}$$

A chord in which both electrodes have identical FTPM values has zero interval distortion; only gradients in mismatch across electrodes disrupt harmonic structure.

Statistical Models. The model set is summarised in Table 4.1. All models are graded LMMs with a random intercept for participant, predicting the signed preference grade (range −5 to +5; Grade = 0 retained for no-preference trials). The models are organised hierarchically around a single narrative. M-Base (inversion only) establishes the dominant root-position preference and serves as the reference against which all other models are compared via AIC. M-Grad (FTPM gradient only) tests the mechanistically appropriate predictor directly: because uniform mismatch preserves interval structure whilst gradients across electrodes destroy it, the inter-electrode FTPM difference is the theoretically motivated predictor rather than raw distortion alone. M-Add (distortion + inversion) controls for the root-position confound additively, testing whether distortion exerts a preference effect beyond chord type - this is the model in which the Pre-ECI distortion effect is detectable. M-Int (distortion × inversion) is the key model: it relaxes the assumption of a common distortion slope across inversion conditions and tests directly whether distortion sensitivity is conditional on chord type. Two group comparison models then examine whether distortion sensitivity differs between groups: M-Grp adds group membership additively, and M-Grp× tests the group-by-distortion interaction. Finally, M-Int$_{\text{bin}}$ is the binary GLLM equivalent of M-Int, fitted on decided trials only (Grade ≠ 0) as a sensitivity check to confirm that results are not driven by the handling of tied responses.

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Table 4.1: **Model set for the chord-level interval distortion analysis (Analysis 1).** All primary models are graded LMMs predicting the signed preference grade (range −5 to +5; Grade = 0 retained for no-preference trials) with a random intercept for participant. Δd: difference in interval distortion between Sound 1 and Sound 2 (Equation 4.9); $\Delta\text{FTPM}_{\text{grad}}$: difference in FTPM gradient (absolute inter-electrode FTPM difference) between the two chords; S1_IsRoot: binary indicator for whether Sound 1 was in root position (1) or first inversion (0); Group: Post-LCI vs Pre-ECI. Models with '×' include both main effects and their interaction. The binary sensitivity check uses a GLLM with a logit link on decided trials only (Grade ≠ 0).

Model	Fixed effects	Role	*Research question*
M-Base	S1_IsRoot	Baseline	*Is preference explained by chord type alone?*
M-Grad	$\Delta\text{FTPM}_{\text{grad}}$	FTPM gradient mechanism	*Does the FTPM gradient between chord electrodes predict preference?*
M-Add	Δd + S1_IsRoot	Distortion + inversion control	*Does distortion predict preference after controlling for chord type?*
M-Int	$\Delta d\times$ S1_IsRoot	Inversion interaction	*Does the distortion effect differ between root-position and inverted chords?*
M-Grp	Δd + Group	Group additive	*Do Post-LCI and Pre-ECI differ in overall distortion sensitivity?*
M-Grp×	$\Delta d\times$ Group	Group interaction	*Does the magnitude of the distortion effect differ between groups?*
Sensitivity check (binary GLLM, decided trials only):			
$\text{M-Int}_{\text{bin}}$	$\Delta d\times$ S1_IsRoot	Binary equivalent of M-Int	*Does the interaction replicate when excluding no-preference trials?*

Analysis 2 - Exploratory Electrode-Factor Associations

This analysis related the electrode-level FIF profiles derived in Analysis 0 to individual factors FTPM, CS, and NR via a structured set of LMMs (M0–M9; Table 4.2). It is explicitly exploratory, constrained by the small number of participants with available factor data (see Table 3.1).

Relating Preference to Individual Factors. LMMs were fitted with preference score as the dependent variable, individual factors as fixed effects, and a random intercept for participant. All predictors were standardised (zero mean, unit variance) prior to model fitting to allow direct comparison of coefficient magnitudes. The model set (Table 4.2) comprised single-predictor models (M1–M3), additive models (M4–M5), interaction models between FTPM and ENI quality (M6–M7), and group comparison models (M8–M9). Degrees of freedom were estimated using the Satterthwaite approximation. It should

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Table 4.2: **Model set for the exploratory electrode-level factor analysis (Analysis 2).** All predictors standardised (zero mean, unit variance). FTPM: frequency-to-place mismatch (oct); CS: current spread; NR: neural responsiveness. Models M8 and M9 were fitted to the pooled sample only. All models include a random intercept for participant.

Model	Scope	Formula (fixed effects)	*Research question*
M0	All	Preference ~ 1	*Is there systematic variability in electrode preference to be explained?*
M1	All	Preference $\sim$ NR	*Does neural responsiveness predict electrode preference?*
M2	All	Preference $\sim$ CS	*Does current spread predict electrode preference?*
M3	All	Preference $\sim$ FTPM	*Does FTPM predict electrode preference?*
M4	All	Preference $\sim$ NR + CS	*Do NR and CS together predict preference beyond each alone?*
M5	All	Preference $\sim$ NR + CS + FTPM	*Do NR, CS, and FTPM each independently predict electrode preference?*
M6	All	Preference $\sim$ FTPM $\times$ NR	*Does the FTPM effect on preference depend on neural responsiveness?*
M7	All	Preference $\sim$ FTPM $\times$ CS	*Does the FTPM effect on preference depend on current spread?*
M8	Pooled	Preference $\sim$ FTPM + Group	*Does FTPM predict preference after controlling for listener group?*
M9	Pooled	Preference $\sim$ FTPM $\times$ Group	*Does the FTPM effect on preference differ between Post-LCI and Pre-ECI?*

be noted that electrode-level observations within a participant are not independent; the LMM random intercept accounts for between-participant differences in overall preference level but does not model within-participant spatial autocorrelation across the electrode array. Pearson correlations reported in the collinearity diagnostics are likewise computed on pooled electrode-level observations and should be interpreted as descriptive summaries rather than formal tests of association.

Group Comparison. Between-group differences in the relationship between FIF shape and FTPM were assessed via models M8 and M9, which included Post-LCI versus Pre-ECI group membership as an additive and interactive fixed effect alongside FTPM, respectively.

Data Completeness. Not all individual factors were available for all participants (Table 3.1). FTPM was available for seven participants, CS and NR for a partially overlapping subset. All three factors were simultaneously available for only two participants,

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constraining M5–M7 to exploratory status. Models fitted on different subsets are not directly comparable via model selection criteria such as AIC.

Collinearity and Electrode Position Confound. Pairwise correlations among predictors (NR-CS, NR-FTPM, CS-FTPM) were computed on the pooled sample prior to model fitting as descriptive summaries of the degree of overlap between predictors. A structural confound exists between FTPM and electrode position: because electrode arrays do not reach the most apical cochlear regions, FTPM tends to be largest at apical electrodes, introducing a systematic gradient that may be collinear with electrode index (defined here as the sequential integer label assigned to each electrode from apex to base, with electrode 1 being the most apical). Within-participant Pearson correlations between FTPM and electrode index were computed to quantify this confound for each participant.

Random Slopes. A random-slope variant of M3 was attempted. Where convergence failed or slope variance was estimated at near zero, the random-intercept model was retained.

4.2.6 Statistical Analysis

All analyses used LMMs or GLLMs following the general modelling framework of Chapter 3. For Analysis 0, group differences were tested via the `GroupCat_ECI` coefficient in each LMM, and FIF profiles were derived using the Bayesian paired-comparison model and compared across groups using LMMs with random slopes. For Analysis 1, graded LMMs predicting the signed grade were the primary analysis; the binary GLLM M-Int$_{\text{bin}}$ served as a sensitivity check on decided trials only. For Analysis 2, electrode-level LMMs predicted preference scores from individual factors, with group comparison restricted to the pooled dataset. The significance threshold was set at $\alpha = .05$ throughout. For the single-predictor models in Analysis 2 (M1–M3), the Holm-Bonferroni correction was applied within this family of three tests. All analyses were conducted in MATLAB R2025a; FIF estimation used the `PairedCompCalc` Python package (Leijon et al., 2019).

4.3 Results

The same CI participants described in Chapter 3 were tested; full demographic details and factor availability are given in Table 3.1. For Analysis 0, all 24 participants (14 Post-LCI, 10 Pre-ECI) with chord preference ratings were included. For Analysis 1, seven participants with complete FTPM data were included: five Post-LCI and two Pre-ECI. For Analysis 2, individual factors were present for only a subset of participants; all three factors were simultaneously available for only three participants (Post-LCI 3, Post-LCI 5, and Pre-ECI 1), constraining the full model M5 to these cases.

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4.3.1 Analysis 0: Full-Sample Perceptual Trends

Electrode-Level Frequency Importance Functions

Do experienced CI users show systematic electrode-level differences in sound quality, and do Post-LCI and Pre-ECI groups differ in the frequency regions associated with higher or lower preference?

For each participant, a FIF was derived from the paired-comparison responses using the Bayesian model described in Section 4.2.5.

Figure 4.7 shows the group-averaged FIF profiles on a common frequency axis. Both groups showed a similar pattern: preference scores were highest in the apical and mid-frequency regions (below approximately 1.5 kHz) and declined progressively towards the basal end of the array. The LMM with a random slope for normalised electrode position confirmed a significant preference gradient ($\hat{\beta} = -3.19$, $p < .001$), with preference declining from apex to base. The quadratic model provided the best fit by AIC, suggesting a flat then declining profile rather than a uniform linear gradient: preference was relatively stable across apical and mid-array electrodes before dropping in the basal third. The categorical frequency region model confirmed this pattern, with mid-array electrodes significantly lower than apical ($\hat{\beta} = -0.75$, $p = .013$) and basal electrodes substantially lower ($\hat{\beta} = -2.18$, $p < .001$).

No significant group difference was observed in either the overall level of preference differentiation (Group: $p = .39$) or the shape of the preference gradient (ElecPos_norm $\times$ Group: $p = .17$), indicating that Post-LCI and Pre-ECI participants showed comparable electrode-level quality profiles.

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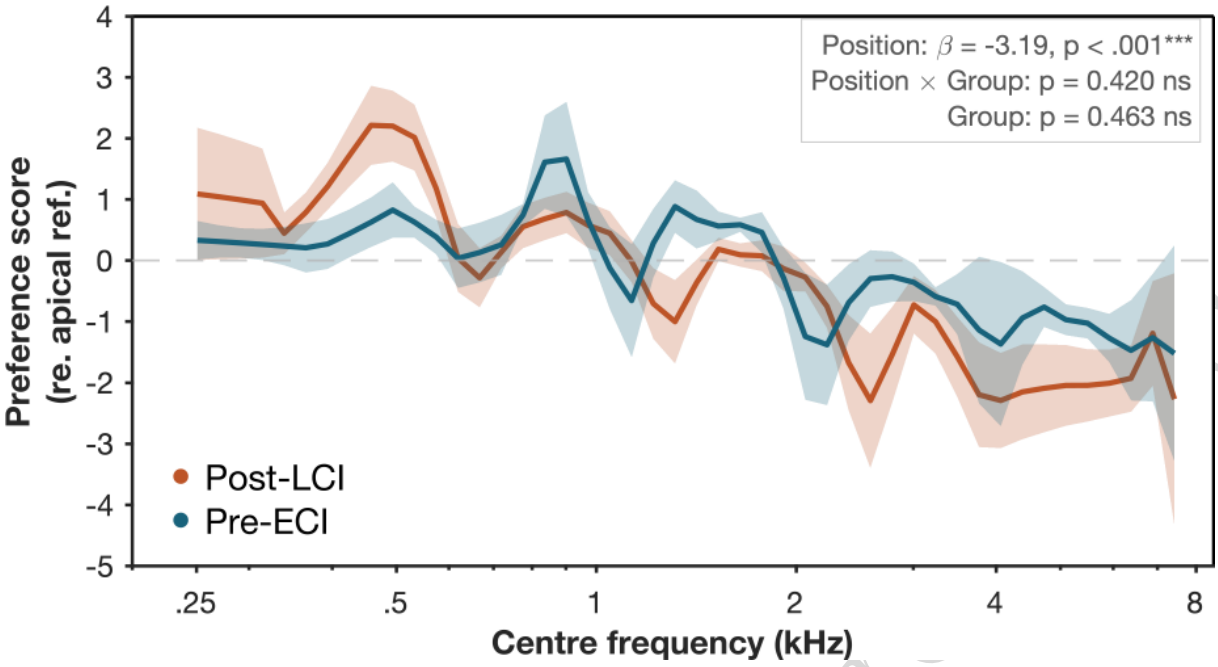


Figure 4.7: **Group-averaged frequency importance functions (Analysis 0).** Group mean FIF profiles interpolated onto a common log-frequency axis. Shaded bands represent ±1 SE across participants. The most apical electrode serves as the Bayesian reference (fixed at zero) for each participant; all preference scores are expressed relative to this reference. Both groups show a comparable pattern of declining preference from apical to basal regions, with no significant group difference in gradient or overall level.

Root-Position Preference

Do experienced CI users show a systematic preference for root-position chords over their inversions, and does this differ between Post-LCI and Pre-ECI groups?

Across the full sample of 24 participants, a strong and consistent preference for root-position chords was observed in both groups. The LMM fitted to Post-LCI participants yielded a significant positive S1_IsRoot coefficient ($\hat{\beta} = +1.16$, $p < .001$), as did the model for Pre-ECI participants ($\hat{\beta} = +1.15$, $p < .001$). The pooled model with a Group × S1_IsRoot interaction showed no significant group difference in root-preference slope ($\hat{\beta}_{\text{interaction}} = -0.007$, $p = .980$) and no significant group main effect ($\hat{\beta}_{\text{ECI}} = +0.43$, $p = .271$), confirming that root-position preference was equivalent in magnitude across Post-LCI and Pre-ECI groups. At the individual level, all but one participant exceeded the 50% chance level, and per-participant OLS regression of signed grade on root-position indicator yielded positive coefficients across the large majority of participants in both groups, with broadly comparable magnitudes. These findings establish that root-position preference is a robust perceptual phenomenon that is not contingent on prior acoustic hearing experience.

No-Preference Rate and Grade Strength

Do Post-LCI and Pre-ECI groups differ in their tendency to express no preference, and in the strength of their graded ratings when they do choose?

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The LMM for no-preference rate showed no significant group difference ($\hat{\beta}_{\text{ECI}} = -0.031$, $p = .619$), indicating that the tendency to respond with no preference was equivalent across groups. The LMM for mean absolute grade on decided trials, however, revealed a significant group effect ($\hat{\beta}_{\text{ECI}} = +0.796$, $p = .043$): Pre-ECI participants expressed stronger preference ratings than Post-LCI participants when they did choose one chord over the other. Figure 4.8 summarises these group-level comparisons.

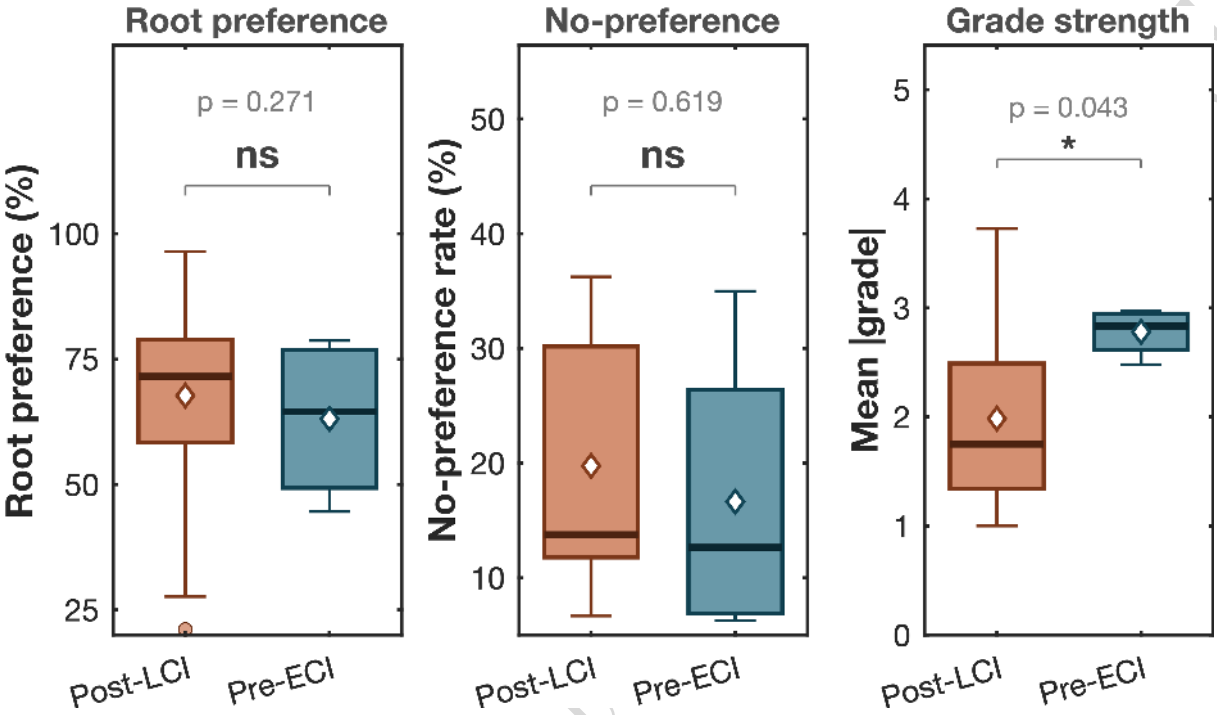


Figure 4.8: **Group comparison of key perceptual metrics (Analysis 0; $N = 24$: 14 Post-LCI, 10 Pre-ECI).** Boxplots show the distribution of per-participant values for root-preference rate (left), no-preference rate (centre), and mean absolute grade on decided trials (right), for Post-LCI and Pre-ECI groups. The diamond marker indicates the group mean. Significance annotations derive from the LMM `GroupCat_ECI` coefficient: ns = not significant; $^{*}p < .05$.

Summary of Analysis 0

The electrode-level FIF profiles revealed a consistent apical-to-basal preference gradient across both groups, with basal electrodes systematically down-weighted relative to apical regions. No significant group difference was observed in FIF shape or overall preference differentiation, indicating that the electrode-level quality landscape is comparable regardless of implantation history. Both Post-LCI and Pre-ECI participants also showed a strong and equivalent root-position preference, with no significant group difference in preference rate, slope, or no-preference rate. The only significant group difference was in preference strength: Pre-ECI users expressed stronger grades when they did choose one chord over the other ($p = .043$). These findings motivate the treatment of root-position preference as a structural confound in the subsequent distortion analyses, and suggest that the full-sample perceptual profile is dominated by an acoustic cue - the structural advantage of root-position chords - rather than by

frequency-to-place factor differences between groups.

4.3.2 Analysis 1: Interval Distortion and Chord Preference

Descriptive Overview

What proportion of trials resulted in selection of the chord with less FTPM-induced interval distortion, and does this differ from chance across the sample?

The seven participants with FTPM data contributed a total of 620 trials (460 Post-LCI, 160 Pre-ECI). Of these, 488 (78.7%) were decided trials (Grade $\neq 0$) and 132 (21.3%) were no-preference trials (Grade $= 0$). The no-preference rate varied considerably across participants (range: 6.2–36.2%), with Post-LCI 5 and Pre-ECI 1 showing the highest rates (36.2% and 35.0%, respectively).

Across all 620 trials, the root-position chord was chosen in 67.4% of decided trials ($p < .001$, binomial test), consistent with the dominant root-position preference observed in Analysis 0. The proportion of decided trials in which the chord with less interval distortion was chosen was 51.4%, not significantly different from chance ($p = .556$, binomial test). A complementary paired t-test on continuous distortion values across the pooled decided sample did not reveal a significant difference between chosen and rejected chords (chosen $M = 0.260$ oct, rejected $M = 0.271$ oct; $t(487) = -1.05$, $p = .292$). Within the Post-LCI subgroup alone, the same test was likewise non-significant (chosen $M = 0.261$ oct, rejected $M = 0.273$ oct; $t(360) = -1.03$, $p = .304$).

The descriptive pattern at the individual level was nonetheless directionally consistent with distortion avoidance in most Post-LCI participants: four of the five showed lower mean distortion for chosen than rejected chords (deltas ranging from +0.006 to +0.074 oct), with Post-LCI 5 as the exception. Among Pre-ECI participants, Pre-ECI 2 showed notably strong binary distortion avoidance (65.3% less-distorted chosen, $p = .011$, binomial), whilst Pre-ECI 1 showed a reversed pattern (38.5%, $p = .126$, ns).

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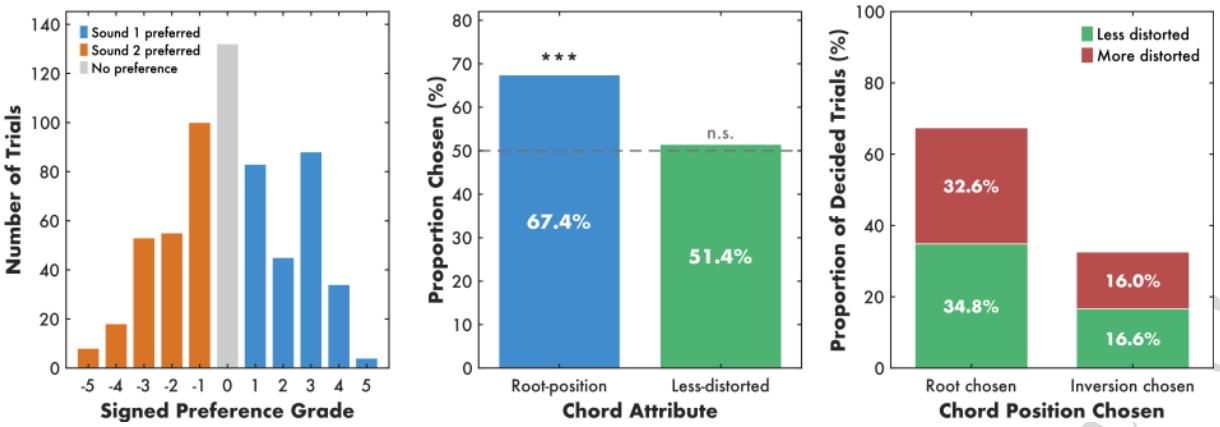


Figure 4.9: **Descriptive overview of chord preference trials (Analysis 1; $N = 7$: 5 Post-LCI, 2 Pre-ECI). Left:** Distribution of signed preference grades (negative = inversion preferred; positive = root preferred). Bars coloured by direction of preference; grey bar indicates grade 0 (no preference). **Centre:** Proportion of decided trials in which the root-position chord and the chord with less interval distortion were chosen, relative to chance (dashed line). Significance assessed by two-tailed binomial test against 50%. **Right:** Stacked breakdown of decided trials by chord position chosen (root vs. inversion) and distortion status (less vs. more distorted), expressed as a percentage of all decided trials.

Root-Position Preference

Is the root-position preference consistent across participants with FTPM data, and does its magnitude vary with group membership?

A root-preference grade was computed for each trial by sign-flipping the signed grade on trials in which the inversion was in the reference position, such that positive values always indicate a preference for the root-position chord. The distribution was markedly right-skewed across the pooled sample, confirming a consistent bias towards root-position chords. Per-participant OLS regression of signed grade on root-position indicator yielded positive coefficients for all seven participants; Post-LCI 5 showed the smallest root-preference coefficient among Post-LCI participants, though it remained positive and significant in the baseline model (M-Base: $\hat{\beta} = +0.51$, $p = .003$). Pre-ECI participants showed root-position coefficients as large as, or larger than, those of most Post-LCI participants, confirming that root-position preference was not contingent on prior musical experience with tonal harmony.

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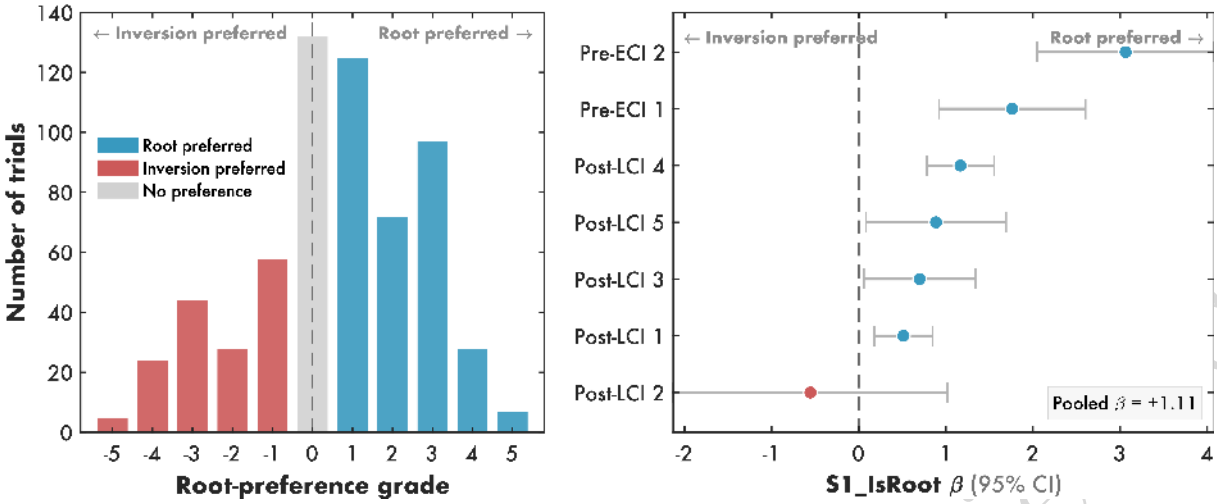


Figure 4.10: **Root-position preference across participants (Analysis 1; $N = 7$). Left:** Distribution of root-preference grades, computed by sign-flipping trials in which the inversion chord was in the reference position. Blue bars indicate trials in which the root-position chord was preferred; red bars indicate trials in which the inversion was preferred; grey bar indicates no preference (grade 0). **Right:** Per-participant forest plot of the root-preference regression coefficient ($\hat{\beta}$), estimated by OLS. Horizontal lines represent 95% confidence intervals.

Distortion Effects: M-Grad, M-Add, and M-Int

Does FTPM-induced interval distortion predict chord preference, and does this effect depend on chord type?

To visualise the distortion-preference relationship in a consistent frame of reference, all trials were recoded relative to the root-position chord: the distortion advantage of the root chord ($\Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$, negative when root is less distorted) and preference for the root chord (positive grade indicating root preferred) were computed for every trial, eliminating the position-assignment coding artefact. Figure 4.11 shows the resulting trial-level scatter with the M-Int simple slope for root-position trials annotated.

The FTPM gradient predictor (M-Grad) was significant in the pooled sample ($\hat{\beta} = -1.53$, $p = .026$) and in the Post-LCI subgroup ($\hat{\beta} = -1.59$, $p = .042$), consistent with the theoretical prediction that gradients in mismatch across chord electrodes - rather than uniform shifts - are the primary driver of harmonic distortion. When chord inversion was added as an additive covariate (M-Add), the distortion effect remained non-significant at the pooled level ($\hat{\beta} = -0.55$, $p = .136$). However, when the Pre-ECI subgroup was analysed in isolation, M-Add revealed a significant additive distortion effect ($\hat{\beta} = -2.62$, $p = .005$), indicating that when inversion is controlled, Pre-ECI participants showed clear distortion avoidance.

The key model M-Int confirmed that the distortion-preference slope differed significantly depending on chord inversion status (Table 4.3; interaction $\hat{\beta} = -3.03$, $p < .001$, Post-LCI subgroup). Simple slopes derived analytically from the model variance-covariance matrix revealed two contrasting patterns: when the root chord was in the reference position, the distortion slope was -2.03 (distortion avoidance), whereas when the inversion was in the reference position, the slope was $+1.00$ (ns, $p = .078$), indi-

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cating that root-position preference dominated regardless of distortion level. The same model fitted to the full pooled sample yielded an equivalent pattern ($\hat{\beta}_{\text{interaction}} = -2.41$, $p = .003$; AIC = 2608.3, best-fitting pooled model). The binary sensitivity check M-Int$_{\text{bin}}$ replicated the interaction on decided trials only ($p < .001$), confirming that the result was not driven by the retention of no-preference trials.

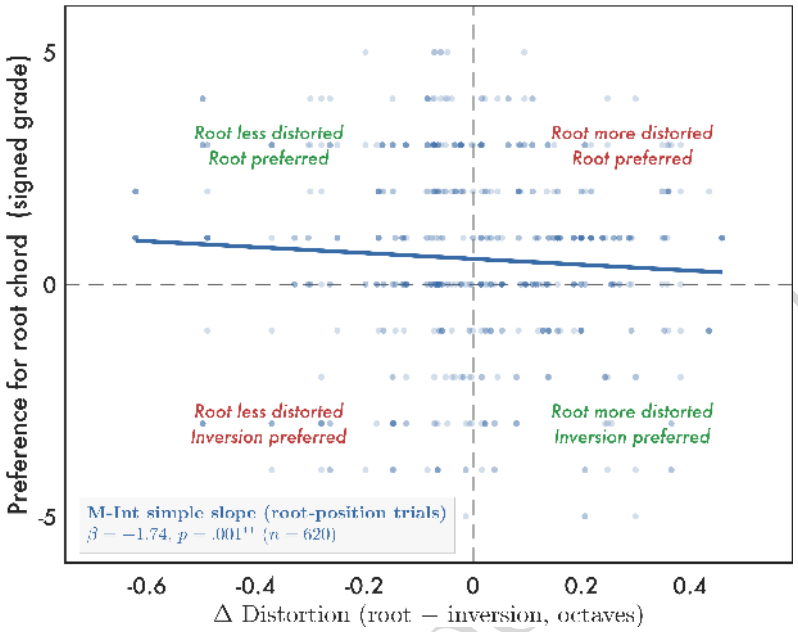


Figure 4.11: **Distortion avoidance in root-position trials (Analysis 1; $N = 7$; 620 trials).** Each point represents one trial, recoded so that the x-axis reflects the distortion advantage of the root-position chord ($\Delta d_{\text{root}} = d_{\text{root}} - d_{\text{inv}}$, negative when root is less distorted) and the y-axis reflects preference for the root chord (positive = root preferred). The OLS regression line is annotated with the M-Int simple slope for root-position trials ($\hat{\beta} = -2.03$, $p < .001$): when the root chord carries less interval distortion than its inversion, participants prefer it more strongly. Green quadrant labels indicate the two quadrants consistent with distortion avoidance (top-left: root less distorted and preferred; bottom-right: root more distorted and not preferred); red labels indicate the opposite pattern. First-inversion trials (not shown) yielded a non-significant near-zero slope ($\hat{\beta} = +1.00$, $p = .078$), indicating that root-position preference dominated regardless of distortion level in those trials.

Table 4.3: Post-LCI key model M-Int: SignedGrade $\sim \Delta d\times$ ChordType + (1|Participant).

Parameter	$\hat{\beta}$	SE	df	p	
Intercept	−0.21	0.34	5.6	.566	
Root-position chord	+0.66	0.17	455.1	<.001	***
Δd	+1.00	0.56	459.7	.078	
$\Delta d\times$ Root-position chord	−3.03	0.85	459.8	<.001	***
Random intercept SD = 0.72; Residual SD = 1.81					
AIC = 1875.4; BIC = 1900.2; LogLik = −931.7					

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Individual-Level Interaction Effects

Is the distortion-avoidance effect consistent across individual Post-LCI participants, and does its magnitude and sign differ between Post-LCI and Pre-ECI listeners?

Table 4.4 and Figure 4.12 report the M-Int interaction term for each participant.

Table 4.4: Individual-level Distortion × Chord-type interaction (M-Int) per participant.

Participant	Group	$\hat{\beta}_{\text{interaction}}$	SE	p	
Post-LCI 1	Post-LCI	−1.38	0.68	.046	*
Post-LCI 2	Post-LCI	−6.06	2.58	.020	*
Post-LCI 3	Post-LCI	−2.84	1.97	.153	
Post-LCI 4	Post-LCI	−11.22	5.80	.058	.
Post-LCI 5	Post-LCI	−1.39	0.80	.086	.
Pre-ECI 1	Pre-ECI	+0.15	3.01	.961	
Pre-ECI 2	Pre-ECI	−3.28	3.73	.391	

Among Post-LCI participants, all five showed negative interaction coefficients, consistent with the group-level pattern. Post-LCI 2 showed the strongest effect ($\hat{\beta} = -6.06$, $p = .020$), with a clear distortion slope reversal between chord-type conditions. A within-chord-type correlation analysis confirmed a significant relationship between electrode-level distortion and preference for root-position chords ($r = 0.76$, $p = .004$) but not for first-inversion chords ($r = 0.28$, $p = .374$). Post-LCI 1 showed a significant interaction ($\hat{\beta} = -1.38$, $p = .046$) and was the only participant for whom the paired t-test on trial-level chosen versus rejected distortion reached significance ($t(93) = -2.47$, $p = .016$). Post-LCI 4 showed a marginally significant interaction ($\hat{\beta} = -11.22$, $p = .058$) despite the smallest trial count and the highest no-preference rate among Post-LCI participants (33.3%). Post-LCI 3 showed a non-significant interaction in the expected direction ($\hat{\beta} = -2.84$, $p = .153$); notably, the FTPM gradient was a stronger predictor than distortion for this participant in the M-Grad model ($\hat{\beta} = -3.00$, $p = .025$). Post-LCI 5 showed a marginal interaction in the expected direction ($\hat{\beta} = -1.39$, $p = .086$), though this participant also showed a reversed binary pattern - chosen chords were on average more distorted than rejected chords ($t(50) = 2.77$, $p = .008$) - warranting caution.

Among Pre-ECI participants, the picture differed qualitatively. Pre-ECI 1 showed no evidence of distortion sensitivity ($\hat{\beta} = +0.15$, $p = .961$), with preference dominated entirely by root-position preference. Pre-ECI 2 showed the strongest binary distortion avoidance of any participant (65.3% of decided trials less-distorted, $p = .011$, binomial) and a significant additive distortion effect once chord type was controlled (M-Add: $\hat{\beta} = -3.88$, $p = .001$). However, the M-Int interaction was not significant for this participant ($\hat{\beta} = -3.28$, $p = .391$), indicating that Pre-ECI 2's distortion avoidance was not conditional on chord type - in contrast to the Post-LCI pattern.

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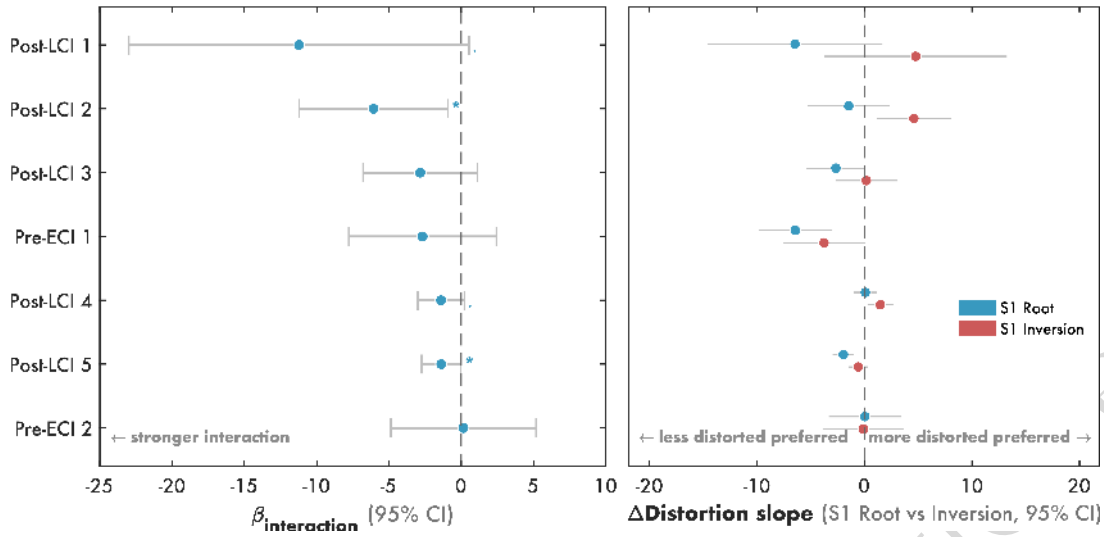


Figure 4.12: **Individual-level Distortion × Chord-type interaction (M-Int; $N = 7$). Left:** Per-participant M-Int interaction coefficient ($\hat{\beta}_{\text{interaction}}$; 95% CI). Negative values indicate that distortion avoidance is stronger in root-position trials than in first-inversion trials. Significance markers: $^*p < .05$; . $p < .10$. **Right:** Simple slopes ($\Delta d\ \hat{\beta}$, 95% CI) per participant, split by chord-type condition. Teal dots: root-position trials; red dots: first-inversion trials.

Group Comparison: M-Grp and M-Grp×

Do Post-LCI and Pre-ECI groups differ in their overall sensitivity to interval distortion (M-Grp), or in the magnitude of the distortion effect on preference (M-Grp×)?

The Distortion × Group interaction in the pooled graded model M-Grp× was not significant ($\hat{\beta} = +0.21$, $p = .828$). The binary sensitivity check M-Int$_{\text{bin}}$ fitted with a group interaction likewise showed no significant group effect ($p = .558$). The Pre-ECI subgroup paired t-test was non-significant ($t(126) = -0.30$, $p = .765$), though this reflects the averaging of Pre-ECI 2's strong avoidance with Pre-ECI 1's null pattern rather than genuine group equivalence.

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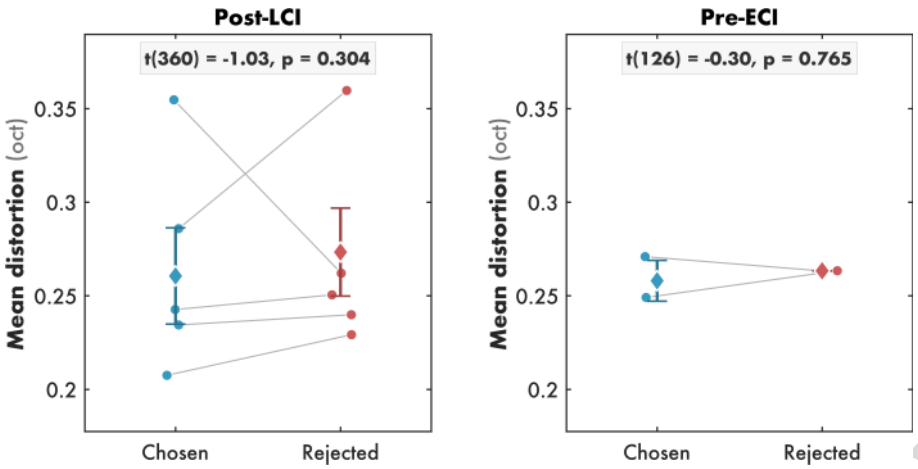


Figure 4.13: **Chosen versus rejected chord distortion by group (Analysis 1).** Per-participant mean distortion for chosen and rejected chords (circles), connected by lines. The diamond marker shows the group mean; whiskers show ±1 SE. Blue: chosen chord; red: rejected chord. **Left:** Post-LCI subgroup ($n = 5$, 361 decided trials). **Right:** Pre-ECI subgroup ($n = 2$, 127 decided trials). Paired t-test results (trial-level) are shown per panel.

Structural Confound Assessment

Root-position and first-inversion chords did not differ systematically in mean distortion across participants. In the pooled Post-LCI data, root-position chords had mean distortion of 0.250 oct versus 0.269 oct for first-inversion chords. Although inversion chords tended to have slightly higher distortion, the difference was small relative to within-condition variance and did not reach significance at the individual level for most participants. M-Int addresses this confound directly by estimating separate distortion slopes for each chord-type condition.

Summary of Analysis 1

Chord quality preference was primarily driven by root-position preference, with participants favouring root-position chords in approximately 67% of decided trials. Interval distortion exerted a secondary influence that was detectable through M-Int in Post-LCI participants: distortion sensitivity manifested in root-position trials ($\hat{\beta} = -2.03$, $p < .001$), but was overwhelmed by root-position preference in first-inversion trials ($\hat{\beta} = +1.00$, ns). M-Int provided the best fit to the Post-LCI subgroup ($p < .001$) and the pooled sample ($p = .003$), and the result was replicated in the binary sensitivity check M-Int$_{\text{bin}}$. All five Post-LCI participants showed negative interaction coefficients (two significant, two marginal, one non-significant in the expected direction). The Pre-ECI pattern was heterogeneous: Pre-ECI 2 showed strong distortion avoidance via M-Add that was not conditional on chord type, whilst Pre-ECI 1 showed no distortion sensitivity under any model.

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4.3.3 Analysis 2: Exploratory Electrode-Factor Associations

The goal of this analysis was to relate the FIF profiles derived in Analysis 0 to individual frequency-to-place factors, in order to quantify the contribution of FTPM, CS, and NR to electrode-level sound quality. This analysis is exploratory, constrained by the small number of participants with available factor data.

Collinearity Diagnostics

Are the three individual factors (NR, CS, FTPM) sufficiently independent to be interpreted separately?

Among the pooled electrode-level observations (Figure 4.14), NR and CS were moderately correlated ($r = -0.44$, $p < .001$), indicating that electrodes with higher neural responsiveness tended to have narrower current spread. NR and FTPM showed a weaker, non-significant correlation ($r = +0.24$, $p = .124$). CS and FTPM showed a marginal negative correlation ($r = -0.24$, $p = .054$). The NR-CS correlation warranted caution in interpreting joint models M4 and M5. It should be noted that these correlations are computed on pooled electrode-level observations across participants; electrode-level data points within a participant are not independent, and these values should therefore be interpreted as descriptive summaries of the degree of overlap between predictors rather than as formal tests of association.

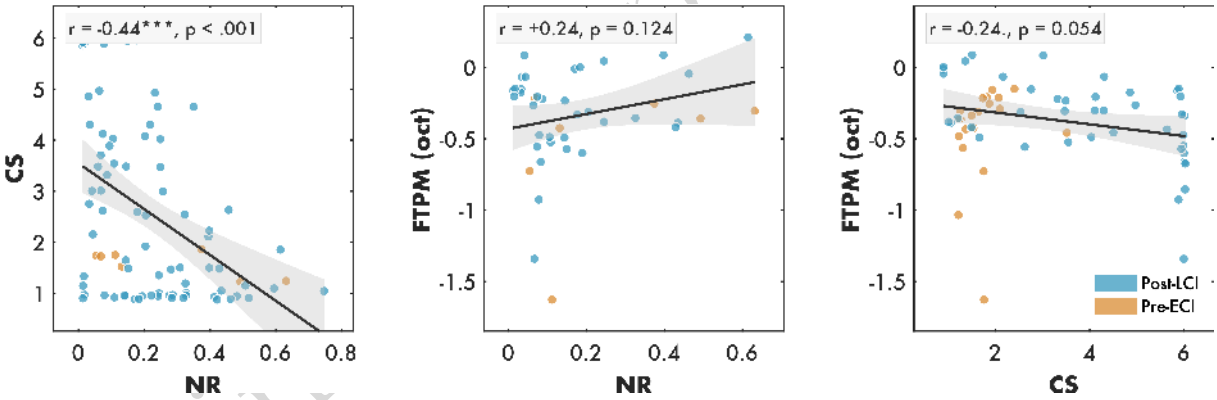


Figure 4.14: **Pairwise collinearity diagnostics for electrode-level factors (exploratory).** Each panel shows electrode-level observations (Post-LCI: blue; Pre-ECI: orange) with a pooled OLS regression line and 95% confidence interval. Pearson r and p-value are annotated per panel. Note: data points within a participant are not independent. **Left:** NR vs CS ($r = -0.44$, $p < .001$). **Centre:** NR vs FTPM ($r = +0.24$, $p = .124$). **Right:** CS vs FTPM ($r = -0.24$, $p = .054$).

Post-LCI: Single-Predictor Models

Does NR (M1), CS (M2), or FTPM (M3) independently predict electrode preference scores in the Post-LCI subgroup?

In the Post-LCI subgroup (Figure 4.15), FTPM was the only significant single predictor of electrode-level preference (M3: $\hat{\beta} = +0.45$, $p = .035$). The positive sign was in the predicted direction: electrodes with less mismatch tended to receive higher preference

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scores. However, FTPM covaried with electrode position in this sample (Section 4.3.3), so the effect may partly reflect a preference gradient along the cochlea. CS showed a marginal trend in the expected direction (M2: $\hat{\beta} = -0.40$, $p = .077$). NR was not significant as a single predictor (M1: $\hat{\beta} = +0.01$, $p = .959$).

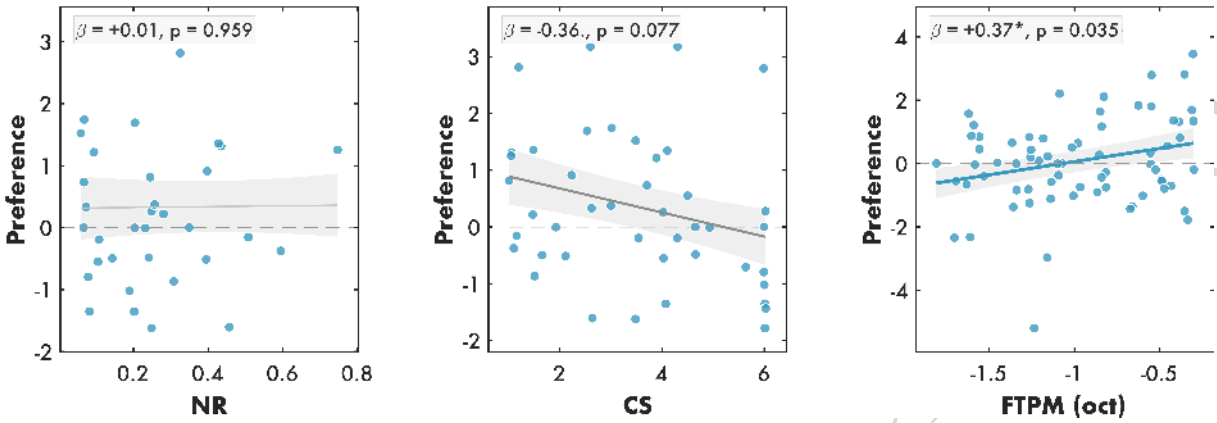


Figure 4.15: **Single-predictor LMM results for the Post-LCI subgroup (exploratory; $n = 5$, 42 electrode observations).** Each panel shows electrode-level preference scores as a function of one factor (NR, CS, FTPM). Dots represent individual electrodes; the line shows the LMM fixed-effect prediction with 95% CI shading. Line weight and colour encode significance: thick blue = significant ($p < .05$); medium grey = marginal ($p < .10$); thin light grey = non-significant. Standardised $\hat{\beta}$ and p-values are annotated per panel.

Post-LCI: Additive and Full Models

Do NR and CS each independently predict electrode preference when entered jointly (M4), and do all three factors contribute independently in the full model (M5)?

When NR and CS were entered jointly (M4; Figure 4.16), CS became significant ($\hat{\beta} = -0.69$, $p = .025$) whilst NR remained non-significant ($\hat{\beta} = -0.34$, $p = .182$). The sign reversal for NR between M1 ($\hat{\beta} = +0.01$) and M4 ($\hat{\beta} = -0.34$) is consistent with suppression arising from the NR-CS collinearity. The full model M5, constrained to the two participants with all three factors available, retained the CS effect ($\hat{\beta} = -0.65$, $p = .037$), whilst FTPM was non-significant ($\hat{\beta} = -0.39$, $p = .467$) and NR marginal ($\hat{\beta} = +0.59$, $p = .118$). The small sample renders M5 exploratory at best.

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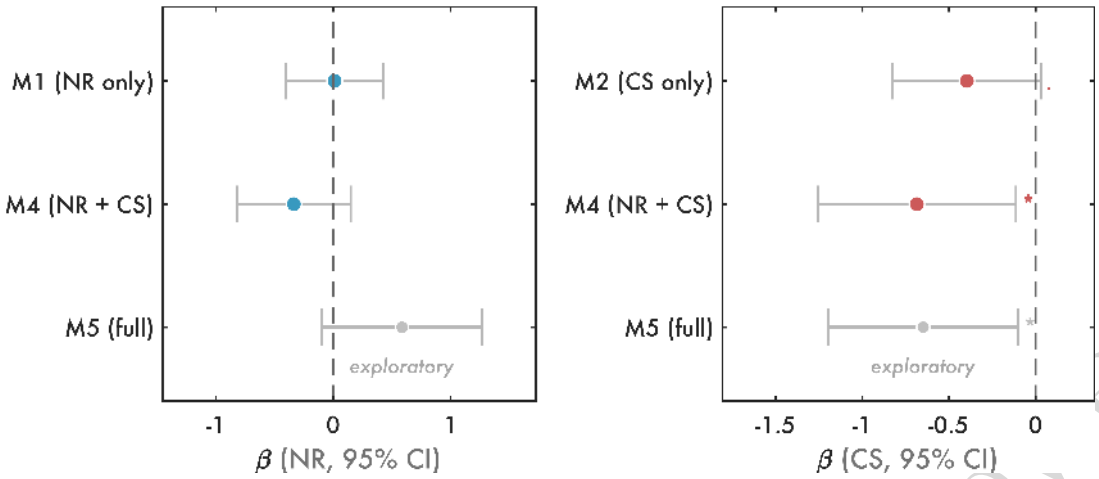


Figure 4.16: **NR and CS coefficients across additive models (Post-LCI; exploratory). Left:** NR $\hat{\beta}$ in M1, M4, and M5. The sign reversal from M1 to M4 reflects suppression driven by NR-CS collinearity. **Right:** CS $\hat{\beta}$ across the same models, showing strengthening once NR variance is partialled out. Error bars show 95% CI. Significance markers: $^*p < .05$; . $p < .10$. The M5 dot is shown in grey to indicate its exploratory status.

Post-LCI: Interaction Models (Exploratory)

Does the FTPM effect on electrode preference depend on local NR (M6) or local CS (M7)?

The FTPM × NR interaction model M6 produced significant coefficients for all terms (NR: $\hat{\beta} = -3.76$, $p = .003$; FTPM: $\hat{\beta} = +6.15$, $p < .001$; interaction: $\hat{\beta} = +6.78$, $p < .001$). The FTPM × CS interaction (M7) was also significant (CS: $\hat{\beta} = +1.46$, $p = .093$; FTPM: $\hat{\beta} = +5.14$, $p = .014$; interaction: $\hat{\beta} = -3.27$, $p = .014$). Both interaction models were fitted on very small samples and are reported as exploratory.

Group Comparison

Does FTPM predict electrode preference after controlling for group membership (M8), and does the FTPM effect differ between Post-LCI and Pre-ECI listeners (M9)?

The group comparison models were fitted to the pooled sample with FTPM data available for both groups. M8 revealed a significant group effect: Post-LCI participants had higher mean electrode-level preference scores than Pre-ECI participants ($\hat{\beta}_{\text{Group, LCI}} = +0.68$, $p = .046$), consistent with greater perceptual differentiation across the electrode array in post-lingually implanted users. The FTPM × Group interaction (M9) was significant ($\hat{\beta} = +0.81$, $p = .043$; Figure 4.17). Decomposing the interaction, the simple slope for Post-LCI was +0.37 (less mismatch associated with higher preference, the predicted direction) and for Pre-ECI was approximately −0.35 (counter to predictions). The opposing slopes explain why the pooled M3 showed a near-zero FTPM effect ($\hat{\beta} = +0.06$, $p = .744$).

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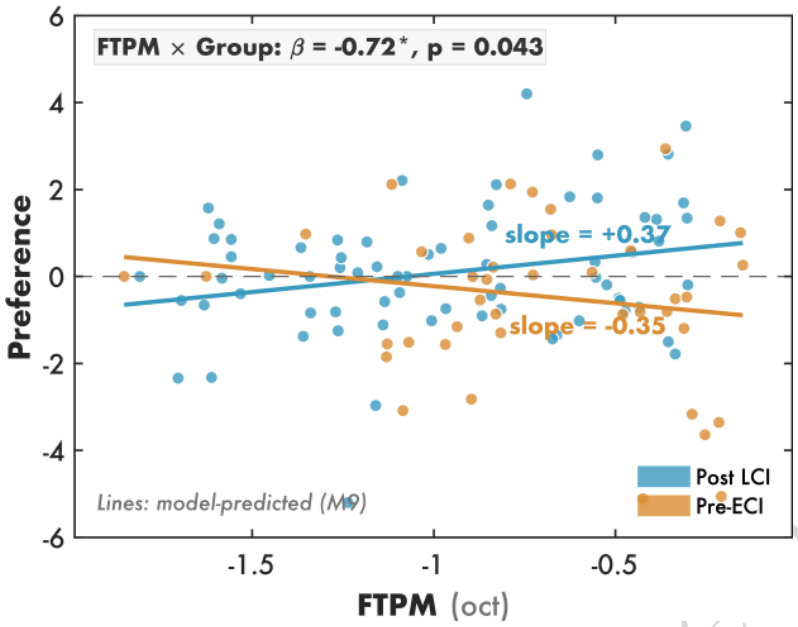


Figure 4.17: **FTPM × Group interaction on electrode preference (M9; exploratory; $N = 7$).** Electrode-level preference scores as a function of FTPM (oct) for Post-LCI (blue) and Pre-ECI (orange) participants. Lines show model-predicted simple slopes from M9. The opposing slopes (+0.37 for Post-LCI, ≈ -0.35 for Pre-ECI) illustrate how the two groups' effects cancel in the pooled model without a group term.

Pre-ECI Subgroup

Do NR (M1), CS (M2), or FTPM (M3) predict electrode preference in the Pre-ECI subgroup?

No single predictor reached significance in the Pre-ECI subgroup (M1: $\hat{\beta} = +0.43$, $p = .593$; M2: $\hat{\beta} = +0.05$, $p = .918$; M3: $\hat{\beta} = -0.31$, $p = .317$; Figure 4.18). In the full model M5, constrained to the single participant with all three factors available, FTPM reached significance ($\hat{\beta} = -1.21$, $p = .043$), but the direction was counter to predictions and the sample size precludes generalisation.

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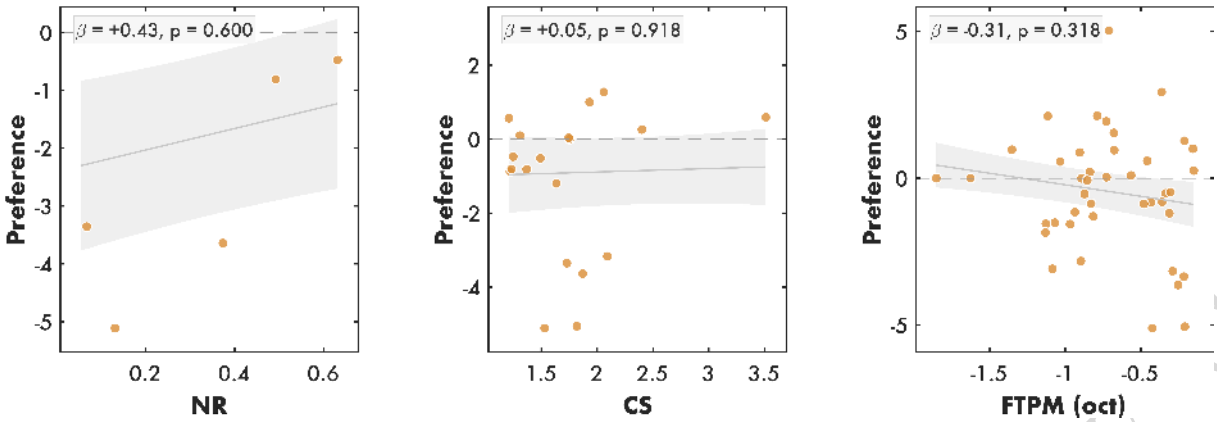


Figure 4.18: **Single-predictor LMM results for the Pre-ECI subgroup (exploratory; $n = 2$).** Layout identical to Figure 4.15. No predictor reached significance in this subgroup. Note: data points are drawn from two participants only; individual profiles are shown in Appendix [X].

Electrode Position Confound

Is FTPM systematically correlated with electrode position along the array, and could this confound the FTPM-preference association observed in M3?

FTPM was significantly correlated with electrode index in the pooled data ($r = +0.25$, $p = .004$). At the individual level, the correlation was significant for four of the seven participants with FTPM data, with Post-LCI 2 as the notable exception showing a negative slope ($r = -0.71$). This structural relationship complicates the interpretation of M3: the positive Post-LCI coefficient ($\hat{\beta} = +0.45$) is in the predicted direction but may be inflated by a preference for basal cochlear regions that coincides with less mismatch.

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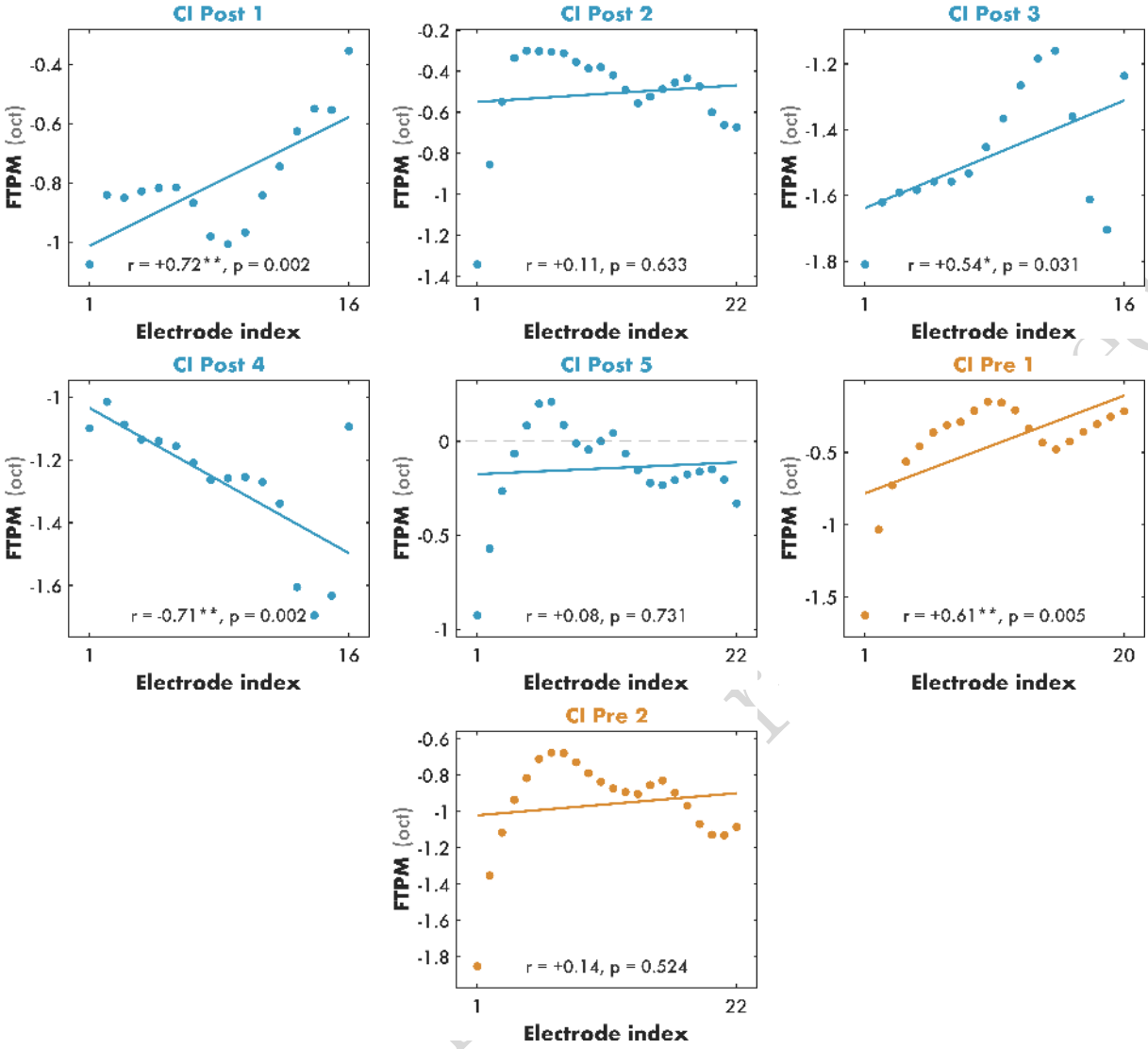


Figure 4.19: **FTPM as a function of electrode index per participant (exploratory; $N = 7$).** Each panel shows electrode-level FTPM values (oct) plotted against electrode index, with an OLS regression line. Pearson r and p-value are annotated per panel; note that electrode-level observations within a participant are not independent. A positive slope indicates that apical electrodes have greater mismatch than basal electrodes.

Random Slopes

Do participants differ systematically in their FTPM-preference slope?

The random-slope variant of M3 estimated a near-zero slope variance in all groups, and the fixed-effect estimate was unchanged from the random-intercept model, indicating either homogeneous FTPM-preference slopes across participants or insufficient power to estimate between-participant slope variability.

Summary of Analysis 2

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The exploratory electrode-level factor analysis yielded a preliminary pattern. CS was the most consistent single predictor: electrodes with narrower current spread tended to be preferred, an effect that reached significance in the joint NR-CS model (M4: $\hat{\beta} = -0.69$, $p = .025$) and was maintained in the full model M5 ($\hat{\beta} = -0.65$, $p = .037$). FTPM showed a significant positive association with preference in the Post-LCI group (M3: $\hat{\beta} = +0.45$, $p = .035$), consistent with the prediction that less mismatch leads to higher preference, though the electrode position confound means this cannot be interpreted unambiguously. The FTPM × Group interaction (M9) was significant ($p = .043$), with opposing FTPM slopes in the two listener groups (Post-LCI $\approx +0.37$; Pre-ECI ≈ -0.35), confirming that the absence of a pooled FTPM effect reflects cancellation rather than a genuine null. The overall group effect on electrode-level preference was significant (M8: $\hat{\beta} = +0.68$, $p = .046$), with Post-LCI participants showing higher mean preference differentiation across the electrode array than Pre-ECI participants, consistent with greater sensitivity to electrode-level spectral quality in users with prior acoustic hearing experience. Interaction models M6 and M7 hinted that the FTPM-preference relationship may be modulated by local NR and CS, but these results are constrained to a small number of participants and must be considered provisional.

4.4 Discussion

This chapter examined whether FTPM, CS, and NR affect chord perception in experienced CI users, and whether those effects differ between Post-LCI and Pre-ECI participants. Three complementary analyses addressed this question at different levels of the paired-comparison task: Analysis 0 characterised perceptual trends across the full sample independently of factor data; Analysis 1 examined chord-level preference between root and inverted versions in relation to FTPM-induced distortion; and Analysis 2 examined electrode-level perceptual weight through FIFs derived from the same task, comparing these weights with all three frequency-to-place factors.

The first sub-goal was to describe the overall pattern of chord preferences, to derive electrode-level FIFs across the full sample, and to examine whether perceptual tendencies differed between Post-LCI and Pre-ECI groups. Analysis 0 revealed a systematic apical-to-basal preference gradient in the FIF profiles that was equivalent across groups, confirmed that root-position preference was equally strong in both groups, and showed that Pre-ECI participants expressed stronger preference grades on decided trials.

The second sub-goal was to determine whether experienced CI users systematically prefer chord versions whose harmonic structure is less distorted by their individual FTPM profile. Analysis 1 confirmed this in part: a detrimental effect of within-chord FTPM-induced distortion was observed, but with different expressions across groups. In Post-LCI users the effect was conditional on chord inversion, whilst in Pre-ECI 2 distortion avoidance was inversion-unconditional.

The third sub-goal was to characterise FIFs across the electrode array and to examine whether electrodes associated with greater absolute FTPM magnitude receive reduced perceptual weight. Analysis 2 confirmed this in the Post-LCI group, with electrodes having larger absolute mismatch associated with lower preference weight.

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The fourth sub-goal extended this to ENI quality: CS showed a clear effect (narrower spread associated with greater electrode preference) whereas NR had no detectable influence. The group comparison across both factor analyses (Analysis 1 PG3a; Analysis 2 M8, M9) further supported differences between Post-LCI and Pre-ECI users, though the qualitative nature of those differences was more nuanced than originally anticipated.

All these points are discussed in turn, beginning with Analysis 0, then Analysis 1, then Analysis 2, and finally both factor analyses conjointly.

4.4.1 Analysis 0: Full-Sample Perceptual Trends

Electrode-Level Frequency Importance Functions The FIF profiles derived from the full sample revealed a systematic preference gradient along the electrode array, with apical and mid-array electrodes receiving higher preference weights than basal electrodes. This gradient was equivalent across Post-LCI and Pre-ECI groups, and the best-fitting model was quadratic, suggesting a relatively flat profile across apical and mid-frequency regions followed by a decline in the basal third of the array. An immediate interpretation of this pattern would be that low-frequency electrodes contribute more to music-related sound quality than high-frequency electrodes, consistent with the literature demonstrating the importance of low-frequency content for sound quality perception in CI users (Roy et al., 2012; Mo et al., 2022; Demirel et al., 2025). However, we consider this interpretation unlikely in the present context, for reasons discussed in detail in Section 4.4.2: because the Bayesian model derives electrode-level weights from paired comparisons in which root-position chords systematically concentrate energy on apical electrodes, the strong root-position preference observed across both groups would inflate the preference weights assigned to those apical electrodes independently of any genuine frequency-region advantage. The apical preference gradient in the FIF profiles is therefore more parsimoniously explained as a consequence of the root-position confound than as evidence for a frequency-dependent perceptual asymmetry. This interpretation is further supported by the absence of a group difference in the gradient: if the apical advantage reflected sensitivity to FTPM-induced distortion - which is typically greatest at apical electrodes - one would expect Post-LCI users to show a different profile from Pre-ECI users, which was not observed.

Root-Position Preference and Grade Strength The dominant finding was that root-position preference was strong, consistent, and equivalent across Post-LCI and Pre-ECI groups. Root-position chords were chosen on approximately two thirds of decided trials in both groups, and the LMM group interaction on root-preference slope was near-zero ($p = .980$). This equivalence is important for two reasons. First, it establishes that root-position preference is not contingent on prior acoustic hearing experience, and rules out accounts of the effect based on stored acoustic harmonic templates. Second, it motivates treating root-position preference as a structural confound in the subsequent distortion analyses rather than as a group-specific perceptual tendency.

The only significant group difference in Analysis 0 was in preference strength: Pre-ECI participants expressed stronger grades on decided trials ($\hat{\beta}_{\text{ECI}} = +0.796$, $p = .043$). This is an unexpected finding. One interpretation is that Pre-ECI users, who have never adapted an acoustic template to the CI signal, are less uncertain about their perceptual

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judgements when they do perceive a difference, and thus commit to stronger ratings. Alternatively, the Pre-ECI group may be more responsive to the specific acoustic cues that drive root-position preference - such as low-frequency energy concentration or interval structure - without the competing influence of FTPM-induced distortion that may introduce ambiguity for Post-LCI users. These interpretations remain speculative in the absence of a larger sample and should be treated accordingly.

The no-preference rate did not differ between groups ($p = .619$), indicating that willingness to express a preference was equivalent, and that the grade strength difference reflects genuine response intensity rather than a lower threshold for committing to a non-neutral response in one group.

4.4.2 Analysis 1: Chord-Level Paired-Comparison

Root-Position Preference Dominates Chord Selection

The most robust finding was the strong preference for root-position chords, observed in six of the seven participants regardless of implantation timing. Participants chose the root-position chord on approximately 67% of decided trials, and the inversion-only model (PG5) outperformed all single-predictor distortion models by a wide margin. Different explanations, mostly driven by the paradigm design itself, may be drawn.

Low-Frequency Content and Music Quality. Root-position chords place the chord fundamental frequency at the lowest frequency, concentrating more energy in the low-frequency region compared to their inverted counterparts, and these root-position chords were judged as more pleasant by CI participants in both groups and in a majority of trials. To contextualise this preference, it is first necessary to consider what role frequency content plays in music perception more broadly. The frequency content of a musical signal has a well-established effect on its perceived quality: Moore and Tan (2003) demonstrated that limiting the bandwidth of music reduces its perceived naturalness, with both low- and high-frequency regions contributing to sound quality judgements. Beyond global bandwidth, different frequency regions carry distinct musical information: low frequencies primarily convey rhythmic and beat-related content, with Hove et al. (2014) showing that the auditory system detects timing deviations more accurately in lower-pitched streams, and Nozaradan et al. (2018) demonstrating at the cortical level that low-frequency sounds selectively boost neural entrainment to the musical beat; high frequencies, by contrast, carry melodic and pitch-related information as well as timbral cues associated with instrument attack transients and upper harmonics (*e.g.*, Moore and Tan, 2003). This frequency-region asymmetry is particularly relevant for CI users: rhythm and beat are among the best-preserved musical attributes in CI listeners, whereas melody recognition and pitch discrimination are severely impaired relative to NH peers (McDermott, 2004). Accordingly, several studies have demonstrated that the availability of low-frequency content influences sound quality ratings in CI users: Roy et al. (2012) showed that bass frequency removal degrades sound quality perception, Mo et al. (2022) showed that increasing gain in low and mid-range frequency bands improved sound quality ratings whereas manipulating gain above 2 kHz had no significant effect, and Demirel et al. (2025) reported that sound quality ratings declined monotonically as the high-pass filter cutoff frequency increased. These findings would suggest a straightforward explanation for the root-position prefer-

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ence observed here: root chords concentrate energy in the low-frequency region, and CI users are known to rely on low-frequency content for sound quality judgements. However, such findings were obtained using real pieces of music, which is not the case in the present study where single, isolated chords were presented. Single chords do not carry temporal information - neither rhythm, beat, nor melody - and the conventional justification for low-frequency superiority in CI music perception therefore does not apply directly to the present paradigm. To our knowledge, this distinction has not been made explicitly in the CI music literature, where low-frequency preference has mainly been attributed to the rhythm and beat advantage without consideration of contexts in which static harmonic information is the primary perceptual target. Alternative mechanism must therefore be considered.

Root Salience and Interval Structure Discriminability. Such root-preference may be accounted for by the difference in interval structure between root-position and inverted chords, and how these intervals interact with the CI mapping of each participant. In root position (*e.g.*, C4-E4-G4), the bass note is separated from the note immediately above by a major third (4 semitones, ≈0.33 octaves), reinforcing its perceptual salience as the harmonic root and supporting a coherent harmonic template match (Parncutt, 1988; Terhardt, 1984); furthermore, the upper two notes are sufficiently separated in frequency to span distinct electrode bands. In first inversion (*e.g.*, E4-G4-C5), the two lowest notes are only a minor third apart (3 semitones, ≈0.25 octaves), which may reduce the distinctiveness of the bass note; more critically, given the coarse frequency resolution of CI processing (Friesen et al., 2001), these two closely spaced notes may fall within the same electrode frequency band and be encoded by the same or adjacent electrodes (Shannon et al., 1995), thereby compromising the harmonic coherence of the chord.

Therefore, the root-position preference is more likely a structural consequence of the paradigm design than a reflection of individual FTPM-related distortion. Root-position chords are acoustically privileged in multiple converging ways - lower spectral energy concentration, better root salience, and a more coherent harmonic template - all of which are independent of the listener's implantation timing history. Moreover, Pre-ECI 1, who has no prior acoustic hearing experience, showed an equally strong preference, arguing against an experience-dependent or FTPM-driven account.

FTPM-Distortion Sensitivity is Conditional on Chord Inversion in Post-LCI Users

A second finding of Analysis 1 was that FTPM-related distortion significantly predicted chord preference in Post-LCI users, but only through an interaction with chord inversion (PG3a: $\hat{\beta}_{\text{interaction}} = -3.03$, $p < .001$). When the chosen chord was in root position, reduced distortion increased preference (simple slope $= -2.03$), consistent with the hypothesis that FTPM-induced spectral distortion degrades perceived chord quality. In contrast, when the chosen chord was in first inversion, the distortion slope was non-significant and positive (+1.00), indicating that root-position preference dominated behaviour regardless of distortion level.

This pattern can be interpreted within a *cue integration framework*, in which listeners combine and weight multiple sources of information to guide perceptual decisions (Ernst

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and Banks, 2002; Knill and Pouget, 2004). Root-position information constitutes a highly salient and reliable cue that receives greater weight in the decision process. When the two cues are misaligned (*e.g.*, when the root-position chord carries greater FTPM-related distortion than the inverted chord), root-position salience dominates and distortion sensitivity is attenuated. When the two cues are aligned (*e.g.*, when the root-position chord carries less distortion), both cues reinforce the same choice and Post-LCI users become sensitive to distortion, modulating their decisions using the rating scale.

The interaction was consistent in direction across all five Post-LCI participants, with negative interaction coefficients ranging from -1.38 to -11.22 (two significant: Post-LCI 1 and Post-LCI 2; two marginal: Post-LCI 4 and Post-LCI 5; one non-significant in the expected direction: Post-LCI 3). Among Pre-ECI participants, the pattern differed qualitatively: Pre-ECI 2 showed a significant additive distortion effect in the graded models once inversion was controlled (PG3: $\hat{\beta} = -3.88$, $p = .001$) but not a significant interaction (G3a: $p = .391$), suggesting that this participant avoided distortion regardless of chord inversion rather than conditionally upon it. Pre-ECI 1 showed no distortion effect under any model specification. This dissociation - inversion-conditional distortion sensitivity in Post-LCI versus inversion-unconditional avoidance in Pre-ECI 2 - may reflect a qualitative difference in the perceptual mechanism: Post-LCI users may require the root-position chord to act as a harmonic anchor before distortion becomes salient, whilst Pre-ECI 2 avoidance appears more global and less contingent on harmonic structure. To our knowledge, this is the first demonstration that FTPM-induced interval distortion affects chord quality perception in CI users.

4.4.3 Analysis 2: FIFs and Frequency-to-Place Factors

This analysis was the original objective of the paradigm developed here: create a task in which individual electrodes could be compared using a music-related signal, and in which electrode-level perceptual weights could be inferred and related to individual frequency-to-place factors.

FTPM Magnitude and the Electrode Position Confound

The electrode-level analysis found that electrodes with higher absolute FTPM tended to receive higher preference scores in the Post-LCI group (M3: $\hat{\beta} = +0.45$, $p = .035$). The direction is consistent with predictions, though the interpretation is complicated by a structural confound between FTPM magnitude and electrode position. In the current sample, FTPM was positively correlated with electrode index ($r = +0.25$, $p = .004$ pooled), meaning that apical electrodes tended to have greater mismatch than basal electrodes. Any preference gradient across the cochlea could therefore inflate the apparent FTPM effect, and the Analysis 2 result should be interpreted with this bias in mind.

CS and NR

CS was a more interpretable predictor in Analysis 2. Electrodes with narrower current spread tended to be preferred (M4: $\hat{\beta} = -0.69$, $p = .025$), an effect that survived in the full model. This is consistent with the idea that more spatially focused stimulation produces a

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clearer spectral representation, which may lead to more distinct encoding of the harmonic structure probed by the paradigm. Crucially, CS is not systematically confounded with electrode position in the same way as FTPM, making it a more interpretable predictor.

NR, by contrast, showed no relationship with electrode-weighted preference (M1: $p = .959$). The absence of an NR effect may be explained by the use of monopolar stimulation, which induces broad CS and makes it difficult to isolate the contribution of NR at individual electrode sites. The null NR result is therefore not surprising in this context, though it remains to be established whether a focused stimulation paradigm would reveal a relationship between NR and electrode preference in this task.

4.4.4 Convergence of Analysis 1 and Analysis 2

The two factor analyses shared a common question - does FTPM orient chord preference? - but addressed it at different levels of analysis. The results were directionally consistent: Analysis 1 found that less interval distortion within each chord was associated with higher preference (interaction model PG3a), and Analysis 2 found that less absolute mismatch was associated with higher electrode-level preference weight (M3: $\hat{\beta} = +0.45$). Both analyses therefore point towards a benefit of reduced mismatch for chord pleasantness perception.

However, the two analyses differ in the strength of their causal inference. Analysis 1 is the more interpretable, because the paired-comparison design naturally controls for electrode position: on each trial, both chords are built from the same electrode pair, and only the interval distortion differs between them. Analysis 2, by contrast, compares preference scores across electrode positions within a participant and is therefore vulnerable to any systematic preference gradient along the array. The FTPM-position confound ($r = +0.25$) means that the Analysis 2 effect could be partly or wholly driven by electrode position rather than FTPM-related distortion, and should therefore be interpreted with greater caution.

4.4.5 Post-LCI Versus Pre-ECI Differences

Pre-ECI 1 showed no distortion sensitivity under any model specification, consistent with the hypothesis that sensitivity to FTPM-related distortion requires a stored spectral template built through prior acoustic hearing experience. Pre-ECI 2 showed strong binary distortion avoidance and a significant additive distortion effect once inversion was controlled, but no inversion-conditional interaction. The formal Distortion × Group interaction was not significant at the pooled level ($p = .828$), reflecting the averaging of these two qualitatively different Pre-ECI profiles rather than evidence for group equivalence.

The pattern of results across the two Pre-ECI participants is more interpretable when considered individually. Pre-ECI 1, who has no prior acoustic hearing experience, showed no distortion sensitivity and preference dominated entirely by inversion status. Pre-ECI 2 showed strong distortion avoidance that was not contingent on chord inversion. This inversion-unconditional avoidance may reflect a more global sensitivity to spectral distortion that does not require the harmonic anchoring provided by root-position chords - consistent with a listener who attends primarily to low-level spectral quality rather than to harmonic template matching.

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The group comparison in Analysis 2 provided cleaner evidence for a group difference. Post-LCI participants showed significantly higher overall electrode-level preference scores than Pre-ECI participants (M8: $\hat{\beta} = +0.68$, $p = .046$), suggesting greater perceptual differentiation across the electrode array. The significant FTPM × Group interaction (M9: $p = .043$) indicated opposing FTPM slopes: positive for Post-LCI users (less absolute FTPM associated with higher preference, the predicted direction) and negative for Pre-ECI users (less absolute FTPM associated with lower preference, counter to predictions). Whether this counter-directional Pre-ECI slope reflects a genuine inverse relationship or a statistical artefact of the electrode position confound in a small Pre-ECI sample ($n = 2$) cannot be determined from the present data.

To our knowledge, this is the first evidence at the electrode level that FTPM impacts CI users differently depending on whether they developed hearing acoustically or electrically.

4.4.6 Limitations

Sample Size. The factor analyses reported here are based on five Post-LCI and two Pre-ECI participants with FTPM data. Whilst the within-participant repeated-measures design provides reasonable statistical power for detecting trial-level effects, the between-participant inferences - particularly the group comparison - rest on a very small sample, with the Pre-ECI subgroup reduced to two participants following the exclusion of one participant with erroneous radiological FTPM estimates. The individual-level heterogeneity within Pre-ECI (one non-responder, one strong responder) further limits the interpretation of group-level models, and additional pre-lingual participants are being recruited. Data collection is ongoing, and the patterns reported here may change as additional participants are tested. The interaction models M6 and M7 in Analysis 2 were fitted on 13 and 22 observations respectively and should be treated as hypothesis-generating rather than confirmatory.

Ecological Validity. One goal was to build a task that would serve as a proxy for how each electrode is used and potentially affected by frequency-to-place factors during music perception. However, as described in Section 4.4.2, the use of single isolated chords limits the interpretability of how CI users perceive music in an ecologically valid sense, because single chords do not carry the temporal information - rhythmic structure, melodic contour, harmonic progression - that characterises real pieces of music. This limitation motivates the development of future paradigms that retain the electrode-level sensitivity of the present approach while incorporating richer musical material.

Task-Related Biases. Several caveats of the present paradigm were discussed in Section 4.4.2 and were partly reflected in the results through the dominant root-position preference, which was largely independent of FTPM-related distortion. A further bias concerns the structure of the comparisons: the paradigm only allows direct comparison between electrodes separated by exactly one octave. Although the Bayesian analysis reconstructs a complete picture of electrode-level preference weights across the array, the direct one-octave constraint may introduce noise into the derived FIFs, visible in some participants as local anomalies at electrodes separated by one, two, and three octaves.

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4.4.7 Clinical Implications

The finding that FTPM likely disrupts harmonicity perception in experienced Post-LCI users has potential clinical relevance. Although the present task does not encompass all aspects of music perception, it suggests that even modest remapping approaches - designed to preserve the harmonicity relationship between allocation frequency and the frequency place effectively stimulated - could improve the perceptual coherence of harmonic material without requiring prolonged adaptation. Such targeted interventions would complement the growing body of evidence supporting anatomy-based fitting strategies (Creff et al., 2024; Jiam et al., 2019; Lambriks et al., 2023c).

The finding that CS predicted electrode-level preference suggests that stimulation strategy may also matter for music perception. Focused stimulation techniques such as tripolar configurations (Bierer, 2010) could improve spectral resolution at the electrode level and potentially benefit chord discrimination; however, the interaction between stimulation focality and NR in this context remains to be established.

4.5 Conclusion

This chapter presented the first investigation of FTPM-induced distortion as a predictor of chord quality perception in CI users, using a paired-comparison paradigm designed to probe electrode-level music perception. Four main findings emerged.

First, root-position chords were strongly preferred over their inversions across all 24 participants irrespective of implantation history ($p = .980$ for group comparison), establishing root-position preference as a structural acoustic phenomenon and motivating its treatment as a confound in subsequent analyses. The sole group difference was in preference strength, with Pre-ECI participants expressing stronger grades on decided trials ($p = .043$).

Second, FTPM-induced interval distortion affected chord preference differently across groups. In Post-LCI users, distortion sensitivity was inversion-conditional, emerging only when root-position and less-distorted cues were aligned, consistent with reliability-weighted cue integration. In Pre-ECI 2, distortion avoidance was inversion-unconditional, suggesting a more global sensitivity that does not require harmonic anchoring; Pre-ECI 1 showed no distortion sensitivity, consistent with the absence of a stored acoustic template.

Third, at the electrode level, both FTPM magnitude and CS predicted preference weights in Post-LCI users, whilst NR did not, likely due to monopolar stimulation. Post-LCI users showed higher electrode-level preference differentiation than Pre-ECI users (M8: $p = .046$), and the significant FTPM $\times$ Group interaction (M9: $p = .043$) confirmed opposing relationships to individual FTPM profiles between groups.

Fourth, these findings converge to suggest that FTPM is a genuine determinant of music quality perception in experienced CI users, and that both frequency remapping and focused stimulation represent plausible clinical intervention targets.

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Postface

The study presented in this chapter investigated whether FTPM, CS, and NR predict chord pleasantness perception in experienced CI users, using a paired-comparison paradigm designed to probe electrode-level music perception. The chapter was strongly experimental in its construction: we developed a completely new paradigm and implemented two separate and independent analyses, tracking the effect of spectral distortions in complementary ways. Whilst some unanticipated biases emerged - most notably the strong root-position preference that dominated the paired-comparison signal - several aspects of the design performed well. Participants consistently appreciated the paired-comparison format and found the continuous pleasantness scale intuitive; the Bayesian Bradley-Terry framework was both computationally efficient and sensitive to electrode-level differences.

These strengths suggest that the paradigm, with targeted modifications, has genuine potential beyond its current implementation. The following paragraphs outline the directions we consider most promising.

The Root-Position Bias: a Structural Limitation. The dominant root-position preference was the most consequential unanticipated finding of this experiment, and it is worth reflecting honestly on whether it can be eliminated in a future design. We hypothesise that the preference may be related to the acoustical structure of the chords - specifically the wider and more discriminable bottom interval of root-position voicing - rather than to individual FTPM profiles. Any paradigm that uses two-note or three-note chords with variable voicing would therefore likely reproduce this bias. At the time of writing, we do not see any obvious alternative that would remove it entirely.

A Fitting Tool for Hearing Aid Frequency-Lowering Algorithms. Whilst FTPM is conventionally considered specific to CI users, one condition directly transposes the phenomenon to another population: HA users for whom a frequency transposition or compression algorithm has been applied to compensate for a cochlear dead region. In both cases, frequency information normally processed at one cochlear location is displaced to another - which is, by definition, FTPM. Adapting the chord paradigm for acoustic stimulation would allow direct measurement of whether these algorithms introduce perceptible harmonic distortion, and whether the degree of distortion predicts music quality ratings in hearing aid users.

To our knowledge, no current clinical tool quantifies the perceptual cost of frequency-lowering at the level of individual spectral intervals. A HA version of the paradigm - using acoustic chord stimuli delivered through the device - could identify the frequency-lowering configuration that minimises perceived distortion for an individual user, providing a perceptually grounded complement to other fitting measures.

Interaural FTPM in Bilateral CI users. Bilateral CI users have two independent electrode arrays, and the FTPM profiles of the two implants are rarely identical: insertion depth, array design, and surgical approach differ between ears, producing different electrode-to-cochlear position relationships on each side. When a chord is presented binaurally, the physical interval between two notes is therefore perceived simultaneously

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through two arrays with different tonotopic mappings, introducing an interaural spectral asymmetry that compounds the within-ear interval distortion measured in the present study.

The techniques and framework developed here could be transposed and adapted to characterise this interaural mismatch. Several experimental variants are conceivable: presenting the same chord to both ears simultaneously to study binaural fusion, splitting the notes of a chord across ears to probe inter-aural interval coherence, or systematically varying the relative FTPM profiles of the two arrays to quantify the perceptual cost of interaural asymmetry independently of within-ear distortion.

Personalised Music Design from Electrode Preference Profiles. The FIFs derived from the present paradigm identify, for each individual, which electrode produce the most and least preferred chord percepts. Rather than using this information only to characterise or remediate distortion, it could be inverted: musical material could be composed or arranged to favour electrodes that attract the highest preference weights for that individual. This would also be a way to address the most fundamental limitation of the present paradigm - the use of isolated static chords that carry no temporal information - by confronting the FIF results with real music. If FIFs identify certain electrode positions as consistently preferred, melodic lines and harmonic progressions could be written to concentrate interval relationships within those positions, effectively designing music around the perceptual topology of each user electrode array. This approach - personalised music composition informed by electrode-level preference data - would represent a fundamentally new model of music accessibility for CI users, moving beyond signal processing interventions that aim to reduce distortion towards creative interventions that work within the constraints of the electrode array rather than against them.

Neural Correlates of FTPM-Driven Distortion. The present paradigm relies entirely on subjective preference judgements, which introduces response bias and limits the interpretability of null results. An objective complement would be to record EEG during the chord task and examine the mismatch negativity (MMN) response to each chord pair. If FTPM-induced distortion is perceptually registered at a pre-attentive level, the MMN amplitude should be correlated with the degree of interval distortion imposed by the participant's individual FTPM profile, providing a physiological validation of the behavioural FIF and a measure that does not depend on explicit pleasantness judgements.

Online Implementation and Large-Scale Normative Data. The most immediate practical extension is an online implementation of the chord paradigm. In contrast to the vowel task described in Chapter 3, the simplicity of the chord stimuli, the brevity of the session, and the efficiency of the Bayesian post-processing make a fully remote deployment more feasible. We are currently developing such a platform. The HA extension described above could be incorporated into the same framework, enabling a unified, device-agnostic tool for measuring frequency-to-place distortion across the full spectrum of users with music hearing impairment.

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Bibliography

Abdulrahman Alahmadi, Fida Alzhrani, Abdulrahman Hagr, and Fahad Alghamdi. Pediatric cochlear implantation: The impact of frequency-to-place mismatch on auditory and speech outcomes. *PLOS ONE*, 20(5):e0333088, 2025. doi: 10.1371/journal.pone.0333088.

George Alexiades, Anandhan Dhanasingh, and Claude Jolly. Method to estimate the complete and two-turn cochlear duct length. *Otology & Neurotology*, 36(5):904–907, 2015. doi: 10.1097/MAO.0000000000000620.

Umut Arslan, Hannah Isber, Barbara Shinn-Cunningham, and Antje Ihlefeld. Low-frequency envelope information contributes more than fine structure to speech intelligibility. *Journal of the Acoustical Society of America*, 151(5):3123–3133, 2022. doi: 10.1121/10.0010474.

Deniz Baskent and Robert V. Shannon. Speech recognition under conditions of frequency-place compression and expansion. *The Journal of the Acoustical Society of America*, 113(4):2064–2076, 2004. doi: 10.1121/1.1558357.

Deniz Başkent, Etienne Gaudrain, Terrin N. Tamati, and Anita Wagner. Perception and psychoacoustics of speech in cochlear implant users. In Anthony T. Cacace, Emile de Kleine, Avril Genene Holt, and Pim van Dijk, editors, *Scientific Foundations of Audiology: Perspectives from Physics, Biology, Modeling, and Medicine*, pages 285–319. Plural Publishing, San Diego, CA, 2016. ISBN 978-1-59756-652-0.

Katelyn A. Berg, Hugh M. Birky, Victoria A. Sevich, Aaron C. Moberly, and Terrin N. Tamati. Sound quality, not speech recognition, explains cochlear implant-related quality of life outcomes. *JASA Express Letters*, 5(10):104401, 2025a. doi: 10.1121/10.0039069.

Katelyn A. Berg, Jack H. Noble, Benoit M. Dawant, Leah W. Sunderhaus, Andrea J. DeFreese, Robert F. Labadie, and René H. Gifford. Cochlear implant electrode placement and music perception. *JAMA Otolaryngology–Head & Neck Surgery*, 151(3):220–227, 2025b. doi: 10.1001/jamaoto.2024.4761.

Katrien Berg, Ishan Bhatt, Sona Bhatt, and Bryan E. Pfingst. Selective deactivation of apical electrodes in cochlear implant users. *Trends in Hearing*, 29:1–18, 2024. doi: 10.1177/23312165241298444.

Julie A. Bierer. Threshold and channel interaction in cochlear implant users: evaluation of the tripolar electrode configuration. *The Journal of the Acoustical Society of America*, 121(3):1642–1653, 2007. doi: 10.1121/1.2436712.

Julie A. Bierer and Leonid Litvak. Reducing channel interaction through cochlear implant programming may improve speech perception. *Trends in Hearing*, 20:1–12, 2016. doi: 10.1177/2331216516653389.

Preprint – not peer-reviewed

Julie Arenberg Bierer. Probing the electrode-neuron interface with focused cochlear implant stimulation. *Trends in Amplification*, 14(2):84–95, 2010. doi: 10.1177/1084713810375249.

Julie Arenberg Bierer and Alexandra D. Nye. Comparisons between detection threshold and loudness perception for individual cochlear implant channels. *Ear and Hearing*, 35 (6):641–651, 2014. doi: 10.1097/AUD.0000000000000058.

Robert C. Bilger, F. Owen Black, Neil T. Hopkinson, Eugene N. Myers, James L. Payne, Noel R. Stenson, Armand Vega, and Robert V. Wolf. Evaluation of subjects presently fitted with implanted auditory prostheses. *Annals of Otology, Rhinology & Laryngology*, 86(Suppl 38):1–176, 1977.

Peter Blamey, Françoise Artieres, Deniz Başkent, François Bergeron, Andy Beynon, Emily Burke, Norbert Dillier, Richard Dowell, Bernard Fraysse, Stéphane Gallégo, et al. Factors affecting auditory performance of postlinguistically deaf adults using cochlear implants: An update with 2251 patients. *Audiology and Neurotology*, 18(1):36–47, 2013a.

Peter Blamey et al. Factors affecting auditory performance of postlinguistically deaf adults using cochlear implants: an update with 2251 patients. *Audiology and Neurotology*, 18 (1):36–47, 2013b. doi: 10.1159/000343189.

Colette Boëx, Maria-Izabel Kos, and Marco Pelizzone. Forward masking in different cochlear implant systems. *The Journal of the Acoustical Society of America*, 114(4): 2058–2065, 2003. doi: 10.1121/1.1610452.

Isabelle Boisvert, Maryam Reis, Anna Au, Robert Cowan, and Richard Dowell. Cochlear implantation outcomes in adults: a scoping review. *PLoS ONE*, 15(5):e0232421, 2020a. doi: 10.1371/journal.pone.0232421.

Isabelle Boisvert, Mridula Reis, Andrea Au, Robert Cowan, and Richard C. Dowell. Cochlear implantation outcomes in adults: A scoping review. *PLOS ONE*, 15(5): e0232421, 2020b.

Thomas Bräcker, Markus Meis, Dirk Dittberner, Tobias Wesarg, Antje Aschendorff, and Roland Laszig. Speech intelligibility with a new fine structure coding strategy for the Med-EL cochlear implant system in adults with residual low-frequency hearing. *HNO*, 57(12):1265–1274, 2009. doi: 10.1007/s00106-009-2024-9.

Louis D. Braida, Nathaniel I. Durlach, Richard P. Lippmann, Betty L. Hicks, William M. Rabinowitz, and Charlotte M. Reed. Hearing aids – a review of past research on linear amplification, amplitude compression, and frequency lowering. *ASHA Monographs*, 19: 1–148, 1979.

Jeroen J. Briaire and Johan H. M. Frijns. Unraveling the electrically evoked compound action potential. *Hearing Research*, 205(1–2):143–156, 2005. doi: 10.1016/j.heares.2005.03.020.

Tim Brochier, François Guérit, John M. Deeks, Charlotte Garcia, Manohar Bance, and Robert P. Carlyon. Evaluating and comparing behavioural and electrophysiological estimates of neural health in cochlear implant users. *Journal of the Association for Research in Otolaryngology*, 22(1):67–80, 2021. doi: 10.1007/s10162-020-00773-0.

Preprint – not peer-reviewed

Carolyn J. Brown, Michelle L. Hughes, Barbara Luk, Paul J. Abbas, Amy Wolaver, and Julie Gervais. The relationship between EAP and EABR thresholds and levels used to program the Nucleus 24 speech processor: data from adults. *Ear and Hearing*, 21(2): 151–163, 2000. doi: 10.1097/00003446-200004000-00006.

Nia S. Brown, Sanjena Venkatesh, Tazheh A. Kavoosi, James O. Onyeukwu, Jason A. Brant, and Alexandra E. Quimby. Speech outcomes of cochlear implantation, from 1983 to present: A systematic review. *Otology & Neurotology*, 46(4):393–404, 2025. doi: 10.1097/MAO.0000000000004468.

Lena Bruns, Dirk Mürbe, and Anja Hahne. Understanding music with cochlear implants. *Scientific Reports*, 6:32026, 2016. doi: 10.1038/srep32026.

Meredith T. Caldwell, Nicole T. Jiam, and Charles J. Limb. Assessment and improvement of sound quality in cochlear implant users. *Laryngoscope Investigative Otolaryngology*, 2(3):119–124, 2017a. doi: 10.1002/lio2.71.

Meredith T. Caldwell, Nicole T. Jiam, and Charles J. Limb. Assessment and improvement of sound quality in cochlear implant users. *Laryngoscope Investigative Otolaryngology*, 2(3):119–124, 2017b. doi: 10.1002/lio2.71.

Michael W. Canfarotta, Margaret T. Dillon, Emily Buss, Harold C. Pillsbury, Kevin D. Brown, and Brendan P. O'Connell. Frequency-to-place mismatch: Characterizing variability and the influence on speech perception outcomes in cochlear implant recipients. *Ear and Hearing*, 41(5):1349–1361, 2020a. doi: 10.1097/AUD.0000000000000864.

Michael W. Canfarotta, Margaret T. Dillon, Emily Buss, Harold C. Pillsbury, Kevin D. Brown, and Brendan P. O'Connell. Frequency-to-Place Mismatch: Characterizing Variability and the Influence on Speech Perception Outcomes in Cochlear Implant Recipients. *Ear and Hearing*, 41(5):1349–1361, October 2020b. ISSN 1538-4667. doi: 10.1097/AUD.0000000000000864. URL https://journals.lww.com/ear-hearing/Fulltext/2020/09000/Frequency_to_Place_Mismatch__Characterizing.27.aspx.

Matthew L. Carlson, Douglas P. Sladen, Richard K. Gurgel, Nicole M. Tombers, Christine M. Lohse, and Colin L. Driscoll. Survey of the american neurotology society on cochlear implantation: Part 1, candidacy assessment and expanding indications. *Otology & Neurotology*, 39(1):e12–e19, 2018.

Robert P. Carlyon and Tobias Goehring. Cochlear implant research and development in the twenty-first century: A critical update. *Journal of the Association for Research in Otolaryngology*, 22:481–508, 2021.

Graeme M. Clark. The multiple-channel cochlear implant: the interface between sound and the central nervous system for hearing, speech, and language in deaf people—a personal perspective. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 361:791–810, 2006.

Leah T. Cohen, Lois M. Richardson, Elizabeth Saunders, and Robert S. C. Cowan. Spatial spread of neural excitation in cochlear implant recipients: Comparison of improved ECAP method and psychophysical forward masking. *Hearing Research*, 179(1–2):72–87, 2003. doi: 10.1016/S0378-5955(03)00096-0.

Preprint – not peer-reviewed

Leah T. Cohen, Elizabeth Saunders, Matthew R. Knight, and Robert S. C. Cowan. Psychophysical measures in patients fitted with Contour and straight Nucleus electrode arrays. *Hearing Research*, 212(1–2):160–175, 2006. doi: 10.1016/j.heares.2005.11.005.

Huw R. Cooper, Marilyn Tait, Robert Carpenter, and Thomas P. Nikolopoulos. Spread of excitation measurements in cochlear implant recipients: clinical applications and limitations. In *10th International Conference on Cochlear Implants and Other Implantable Auditory Technologies*, San Diego, CA, 2008. Conference presentation. See also manufacturer-integrated SOE protocols in Cochlear NRT software.

Jason Cramer, Ho-Hsiang Wu, Justin Salamon, and Juan Pablo Bello. Look, listen, and learn more: design choices for deep audio embeddings. pages 3852–3856, 2019. doi: 10.1109/ICASSP.2019.8682475.

Gwenaelle Creff, Nicolas Bernard-Le Liboux, Paul Coudert, Hermine Bourdon, Vincent Pean, Nicolas Wallaert, Cassandre Lambert, and Benoit Godey. Tonotopic and default frequency fitting for music perception in cochlear implant recipients: A randomized clinical trial. *JAMA Otolaryngology–Head & Neck Surgery*, 150(11):960–968, 2024. doi: 10.1001/jamaoto.2024.2895.

Naomi B. H. Croghan, Sarah I. Duran, and Zachary M. Smith. Re-examining the relationship between number of cochlear implant channels and maximal speech intelligibility. *The Journal of the Acoustical Society of America*, 142(6):EL537–EL543, 2017. doi: 10.1121/1.5016044.

Alison DeFreese, Kendra Berg, Jack Noble, Benoit Dawant, Jourdan Holder, et al. Frequency-to-place mismatch and cochlear implant outcomes by electrode type. *JAMA Otolaryngology–Head & Neck Surgery*, 151(2):116–124, 2025. doi: 10.1001/jamaoto.2024.4136.

Andrea DeFreese, Eve Kann, J. Thomas Roland, Mario A. Svirsky, and David M. Landsberger. Frequency-to-place mismatch and speech recognition in cochlear implant users. *Ear & Hearing*, 45(2):330–341, 2024. doi: 10.1097/AUD.0000000000001438.

Ayşenur Karaman Demirel, Ahmet Alperen Akbulut, Ayşe Ayça Çiprut, and Nilüfer Bal. Exploring the link between sound quality perception, music perception, music engagement, and quality of life in cochlear implant recipients. *Audiology Research*, 15 (4):94, 2025. doi: 10.3390/audiolres15040094.

Laura Dessard, Guillaume Gersdorff, Nicola Ivanovik, Masoud Zoca-Assadi, Peter Nopp, Séverine Camby, and Philippe P. Lefebvre. Cochlear implant: analysis of the frequency-to-place mismatch with the table-based software OTOPLAN® and its influence on hearing performance. *Audiology and Neurotology*, 29(3):239–245, 2024.

Lindsay DeVries and Jeroen G. Arenberg. Current focusing to reduce channel interaction for distant electrodes in cochlear implant programs. *Trends in Hearing*, 22: 2331216518813811, 2018. doi: 10.1177/2331216518813811.

Anandhan Dhanasingh and Claude Jolly. An overview of cochlear implant electrode array designs. *Hearing Research*, 356:93–103, 2017. doi: 10.1016/j.heares.2017.10.005.

Preprint - not peer-reviewed

Norbert Dillier, Wai Kong Lai, Brigitta Almqvist, Carolin Frohne, Joachim Müller-Deile, Martin Stecker, and Ernst von Wallenberg. Measurement of the electrically evoked compound action potential via a neural response telemetry system. *Annals of Otology, Rhinology & Laryngology*, 111(5):407–414, 2002. doi: 10.1177/000348940211100505.

Margaret T. Dillon, Michael W. Canfarotta, Emily Buss, and Benjamin P. O'Connell. Comparison of speech recognition with an organ of Corti versus spiral ganglion frequency-to-place function in place-based mapping of cochlear implant and electric-acoustic stimulation devices. *Otology & Neurotology*, 42(5):721–725, 2021. doi: 10.1097/MAO.0000000000003047.

Mishaela DiNino and Julie G. Arenberg. Effects of age and cochlear implantation on spectrally cued speech categorization. *Journal of Speech, Language, and Hearing Research*, 63(8):2425–2440, 2020. doi: 10.1044/2020_JSLHR-19-00127.

André Djourno and Charles Eyriès. Prothèse auditive par excitation électrique à distance du nerf sensoriel à l'aide d'un bobinage inclus à demeure. *La Presse Médicale*, 65:1417, 1957.

Michael F. Dorman et al. Cochlear implant sound quality. *Journal of Speech, Language, and Hearing Research*, 2025. doi: 10.1044/2024_JSLHR-23-00678. TODO: Complete author list and volume/page information.

Ward R Drennan and Jay T Rubinstein. Music perception in cochlear implant users and its relationship with psychophysical capabilities. *Journal of Rehabilitation Research & Development*, 45(5):779–789, 2008. doi: 10.1682/JRRD.2007.08.0118.

Ward R Drennan, Jacob J Oleson, Kate Gfeller, Jillian Crosson, Virginia D Driscoll, Jong Ho Won, Elizabeth S Anderson, and Jay T Rubinstein. Clinical evaluation of music perception, appraisal and experience in cochlear implant users. *International Journal of Audiology*, 54(2):114–123, 2015. doi: 10.3109/14992027.2014.948219.

Michael O. Ernst and Martin S. Banks. Humans integrate visual and haptic information in a statistically optimal fashion. *Nature*, 415(6870):429–433, 2002. doi: 10.1038/415429a.

Andrew Faulkner. Potential benefits from deeply inserted cochlear implant electrodes. *International Journal of Audiology*, 50(S2):S52–S59, 2011. doi: 10.3109/14992027.2010.533704.

Jose N. Fayad and Fred H. Linthicum, Jr. Multichannel cochlear implants: relation of histopathology to performance. *The Laryngoscope*, 116(8):1310–1320, 2006. doi: 10.1097/01.mlg.0000227176.09500.28.

C. C. Finley, T. A. Holden, L. K. Holden, B. R. Whiting, R. A. Chole, G. J. Neely, T. E. Hullar, and M. W. Skinner. Role of electrode placement as a contributor to variability in cochlear implant outcomes. *Otology & Neurotology*, 29(7):920–928, 2008. doi: 10.1097/MAO.0b013e318184f492.

Jill B. Firszt, Laura K. Holden, Margaret W. Skinner, Emily A. Tobey, Amy Peterson, Wolfgang Gaggl, Christina L. Runge-Samuelson, and P. Ashley Wackym. Recognition of speech presented at soft to loud levels by adult cochlear implant recipients of three cochlear implant systems. *Ear & Hearing*, 25(4):375–387, 2004. doi: 10.1097/01.AUD.0000134552.22205.AA.

Preprint - not peer-reviewed

Tom Francart, Jan Brokx, and Jan Wouters. Sensitivity to interaural time differences with combined cochlear implant and acoustic stimulation. *Journal of the Association for Research in Otolaryngology*, 12(6):733–745, 2011. doi: 10.1007/s10162-011-0289-0.

Lindsey M Friesen, Robert V Shannon, Deniz Baskent, and Xiaosong Wang. Speech recognition in noise as a function of the number of spectral channels: Comparison of acoustic hearing and cochlear implants. *Journal of the Acoustical Society of America*, 110(2):1150–1163, 2001. doi: 10.1121/1.1381538.

Johan H. M. Frijns, Jeroen J. Briaire, Jan A. P. M. de Laat, and J. J. Grote. Initial evaluation of the Clarion CII cochlear implant: speech perception and neural response imaging. *Ear and Hearing*, 23(3):184–197, 2002. See also Briaire & Frijns (2005) for spread-of-excitation methodology.

Qian-Jie Fu and John J. Galvin. Perceptual learning following changes in the frequency-to-electrode assignment with the Nucleus-22 cochlear implant. *The Journal of the Acoustical Society of America*, 113(1):351–357, 2003. doi: 10.1121/1.1526428.

Qian-Jie Fu and John J. Galvin. Perceptual learning and auditory training in cochlear implant recipients. *Trends in Amplification*, 11(3):193–205, 2007a. doi: 10.1177/1084713807301379.

Qian-Jie Fu and John J Galvin. Perceptual learning and auditory training in cochlear implant recipients. *Trends in Amplification*, 11(4):193–205, 2007b. doi: 10.1177/1084713807301379.

Qian-Jie Fu and John J. Galvin. Perceptual learning and auditory training in cochlear implant recipients. *Trends in Amplification*, 11(3):193–205, 2007c. doi: 10.1177/1084713807301379.

Christina Fuller, Deniz Başkent, and Rolien Free. Early deafened, late implanted cochlear implant users appreciate music more than and identify music as well as postlingual users. *Frontiers in Neuroscience*, 13:1050, 2019. doi: 10.3389/fnins.2019.01050.

Christopher D Fuller, John J Galvin, Bert Maat, Deniz Båskent, and Rolien H Free. Comparison of two music training approaches on music and speech perception in cochlear implant users. *Trends in Hearing*, 22:2331216518765379, 2018. doi: 10.1177/2331216518765379.

John J. Galvin, Qian-Jie Fu, and Geraldine Nogaki. Melodic contour identification by cochlear implant listeners. *Ear and Hearing*, 28(3):302–319, 2007. doi: 10.1097/01.aud.0000261689.35445.20.

John J Galvin, Elizabeth Eskridge, and Qian-Jie Fu. Deeper cochlear implant electrode insertion angle improves detection of musical sound quality deterioration related to bass frequency removal. *Otology & Neurotology*, 37(2):e92–e98, 2016. doi: 10.1097/MAO.0000000000000941. CHECK: verify author list and page range.

Bruce J. Gantz, Gregg G. Woodworth, John F. Knutson, Paul J. Abbas, and Richard S. Tyler. Multivariate predictors of audiological success with multichannel cochlear implants. *Annals of Otology, Rhinology & Laryngology*, 102(12):909–916, 1993. doi: 10.1177/000348949310201201.

Preprint – not peer-reviewed

Charlotte Garcia, Tobias Goehring, Stefano Cosentino, Richard E. Turner, John M. Deeks, Tim Brochier, Taren Rughooputh, Manohar Bance, and Robert P. Carlyon. The panoramic ECAP method: Estimating patient-specific patterns of current spread and neural health in cochlear implant users. *Journal of the Association for Research in Otolaryngology*, 22(5):567–589, 2021. doi: 10.1007/s10162-021-00795-2.

Ann E. Geers, Christine M. Mitchell, Andrea Warner-Czyz, Nae-Yuh Wang, and Laurie S. Eisenberg. Early sign language exposure and cochlear implantation benefits. *Pediatrics*, 140(1):e20163489, 2017. doi: 10.1542/peds.2016-3489.

Kate Gfeller, Christopher Turner, Jacob Oleson, Xiuhua Zhang, Bruce Gantz, Robin Froman, and Carol Olszewski. Accuracy of cochlear implant recipients on pitch perception, melody recognition, and speech reception in noise. *Ear and Hearing*, 28(3):412–423, 2007. doi: 10.1097/AUD.0b013e3180479318. CHECK: confirm this is the intended paper.

Kate Gfeller, Jacob Oleson, John F Knutson, Patrick Breheny, Virginia Driscoll, and Carol Olszewski. Multivariate predictors of music perception and appraisal by adult cochlear implant users. *Journal of the American Academy of Audiology*, 19(2):120–134, 2008. doi: 10.3766/jaaa.19.2.3.

Kate Gfeller, Dingfeng Jiang, Jacob J. Oleson, Virginia Driscoll, and John F. Knutson. Temporal stability of music perception and appraisal scores of adult cochlear implant recipients. *Journal of the American Academy of Audiology*, 21(1):28–34, 2010. doi: 10.3766/jaaa.21.1.4.

Marcel R. Giezen, Paola Escudero, and Anne Baker. Use of acoustic cues by children with cochlear implants. *Journal of Speech, Language, and Hearing Research*, 53(6): 1440–1457, 2010. doi: 10.1044/1092-4388(2010/09-0107).

René H. Gifford, Michael F. Dorman, Jon K. Shallop, and Sarah A. Sydlowski. Evidence for the expansion of adult cochlear implant candidacy. *Ear and Hearing*, 31(2):186–194, 2010.

René H. Gifford, Jourdan T. Holder, James J. Grisel, Elizabeth L. Perkins, and Robert F. Labadie. Insertion depth and cochlear implant speech recognition outcomes: A comparative study of 28- and 31.5-mm lateral wall arrays. *Otology & Neurotology*, 43(2): e191–e197, 2022. doi: 10.1097/MAO.0000000000003397.

Melanie L Gilbert, Charles J Limb, Eunice K Kim, Mickael L D Deroche, and Rebecca M Lewis. Perception of extreme musical dissonance in cochlear implant users using a novel listening task. *Frontiers in Audiology and Otology*, 3:1554784, 2025. doi: 10.3389/fauot.2025.1554784.

Tobias Goehring, Alexander W Archer-Boyd, John M Deeks, Julie G Arenberg, and Robert P Carlyon. The effect of increased channel interaction on speech perception with cochlear implants. *Scientific Reports*, 11:10424, 2021. doi: 10.1038/s41598-021-89932-8.

Benjamin Goudey, Kerrie Plant, Isabell Kiral, Antonio Jimeno-Yepes, Annalisa Swan, Manoj Gambhir, Andreas Büchner, Eugen Kludt, Robert H. Eikelboom, Cathy Sucher, René H. Gifford, Riaan Rottier, and Hamideh Anjomshoa. A multicenter analysis of

Preprint - not peer-reviewed

factors associated with hearing outcome for 2,735 adults with cochlear implants. *Trends in Hearing*, 25:23312165211037525, 2021. doi: 10.1177/23312165211037525.

Matthew J. Goupell, Christine Stoelb, Alan Kan, and Ruth Y. Litovsky. Effect of mismatched place-of-stimulation on the salience of binaural cues in conditions that simulate bilateral cochlear-implant listening. *The Journal of the Acoustical Society of America*, 133(4):2272–2287, 2013. doi: 10.1121/1.4792936.

Matthew J. Goupell, Olga A. Stakhovskaya, and Joshua G. W. Bernstein. Contralateral interference caused by binaurally presented competing speech in adult bilateral cochlear-implant users. *Ear and Hearing*, 38(1):e110–e123, 2016. doi: 10.1097/AUD.0000000000000357.

Matthew J. Goupell, Stefano Cosentino, Olga A. Stakhovskaya, and Joshua G. W. Bernstein. Interaural pitch-discrimination range effects for bilateral and single-sided-deafness cochlear-implant users. *Journal of the Association for Research in Otolaryngology*, 20 (2):187–203, 2019. doi: 10.1007/s10162-018-00707-x.

Donald D. Greenwood. A cochlear frequency-position function for several species — 29 years later. *The Journal of the Acoustical Society of America*, 87(6):2592–2605, 1990. doi: 10.1121/1.399052.

James D. Harnsberger, Mario A. Svirsky, Adam R. Kaiser, David B. Pisoni, Richard Wright, and Ted A. Meyer. Perceptual "vowel spaces" of cochlear implant users: implications for the study of auditory adaptation to spectral shift. *Journal of the Acoustical Society of America*, 109(5):2135–2145, 2001.

Shawn Harrison, Mehdi Abouzari, Jordan Pang, and Fan-Gang Zeng. Electrocochleography and cognition are important predictors of speech perception outcomes in noise for cochlear implant recipients. *Scientific Reports*, 12:3083, 2022. doi: 10.1038/s41598-022-07175-7.

Shuman He, Jeffrey Skidmore, Brittany Koch, Monita Chatterjee, Brittany L. Carter, and Yi Yuan. The relationship between cochlear implant speech perception outcomes and electrophysiological measures of the electrically evoked compound action potential. *Ear and Hearing*, 45(2):371–384, 2024. doi: 10.1097/AUD.0000000000001289.

James M. Hillenbrand, Laura A. Getty, Michael J. Clark, and Kimberlee Wheeler. Acoustic characteristics of American English vowels. *The Journal of the Acoustical Society of America*, 97(5):3099–3111, 1995. doi: 10.1121/1.411872.

Raul Hinojosa and Margaret Marion. Histopathology of profound sensorineural deafness. *Annals of the New York Academy of Sciences*, 405:459–484, 1983. doi: 10.1111/j.1749-6632.1983.tb31662.x.

Anke Hirschfelder, Silke Gräbel, and Heidi Olze. The impact of cochlear implantation on quality of life: The role of audiologic performance and variables. *Otolaryngology–Head and Neck Surgery*, 138(3):357–362, 2008. doi: 10.1016/j.otohns.2007.10.019.

I. J. Hochmair-Desoyer and E. S. Hochmair. An eight channel scala tympani electrode for auditory prostheses. *IEEE Transactions on Biomedical Engineering*, BME-27(1): 44–50, 1980. doi: 10.1109/TBME.1980.326854.

Preprint – not peer-reviewed

Laura K. Holden, Charles C. Finley, Jill B. Firszt, Timothy A. Holden, Christine Brenner, Lisa G. Potts, Beverly D. Gotter, Shelly S. Vanderhoof, Karen Mispagel, Gina Heydebrand, and Margaret W. Skinner. Factors affecting open-set word recognition in adults with cochlear implants. *Ear and Hearing*, 34(3):342–360, 2013a. doi: 10.1097/AUD.0b013e3182741AA0.

Laura K Holden, Charles C Finley, Jill B Firszt, et al. Factors affecting open-set word recognition in adults with cochlear implants. *Ear and Hearing*, 34(3):342–360, 2013b. doi: 10.1097/AUD.0b013e3182741aa7.

William F. House and Jack Urban. Long term results of electrode implantation and electronic stimulation of the cochlea in man. *Annals of Otology, Rhinology & Laryngology*, 82(4):504–517, 1973. doi: 10.1177/000348947308200408.

Michael J. Hove, Céline Marie, Ian C. Bruce, and Laurel J. Trainor. Superior time perception for lower musical pitch explains why bass-ranged instruments lay down musical rhythms. *Proceedings of the National Academy of Sciences*, 111(28):10383–10388, 2014. doi: 10.1073/pnas.1402039111.

Sarah E. Hughes, Holly A. Hutchings, Frances L. Rapport, Catherine M. McMahon, and Isabelle Boisvert. Social connectedness and perceived listening effort in adult cochlear implant users: A grounded theory to establish content validity for a new patient-reported outcome measure. *Ear and Hearing*, 39(5):922–934, 2018. doi: 10.1097/AUD.0000000000000553.

David Huron. *Sweet Anticipation: Music and the Psychology of Expectation*. MIT Press, Cambridge, MA, 2006. ISBN 9780262083454.

Krista L. Hyde and Isabelle Peretz. Brains that are out of tune but in time. *Psychological Science*, 15(5):356–360, 2004. doi: 10.1111/j.0956-7976.2004.00683.x.

ITU-R – Radiocommunication Sector of ITU. BS.1534-3: Method for the subjective assessment of intermediate quality level of audio systems. Recommendation BS.1534-3, International Telecommunication Union, 2014.

Yannick Jadoul, Bill Thompson, and Bart de Boer. Introducing Parselmouth: A Python interface to Praat. *Journal of Phonetics*, 71:1–15, 2018. doi: 10.1016/j.wocn.2018.07.001.

Christopher J. James, Cyrine Karoui, Marie-Lise Laborde, Benoit Lepage, Corinne Molinier, Joël Tartari, Olivier Deguine, and Bernard Fraysse. Early sentence recognition in adult cochlear implant users. *Ear and Hearing*, 40(4):905–917, 2019. doi: 10.1097/AUD.0000000000000666.

Nicole T. Jiam, Melanie Gilbert, Daniel Cooke, Patpong Jiradejvong, Karen Barrett, Meredith Caldwell, and Charles J. Limb. Association between flat-panel computed tomographic imaging-guided place-pitch mapping and speech and pitch perception in cochlear implant users. *JAMA Otolaryngology–Head & Neck Surgery*, 145(2):109–116, 2019. doi: 10.1001/jamaoto.2018.3096.

Carolyn N. Jolly, Francis A. Spelman, and Ben M. Clopton. Quadrupolar stimulation for cochlear prostheses: Modeling and experimental data. *IEEE Transactions on Biomedical Engineering*, 43(8):857–865, 1996. doi: 10.1109/10.508549.

Preprint – not peer-reviewed

Ruben K. Kalkman, Jeroen J. Briaire, and Johan H. M. Frijns. Stimulation strategies and electrode design in computational models of the electrically stimulated cochlea: an overview of existing literature. *Network: Computation in Neural Systems*, 25(1–2): 1–14, 2014. doi: 10.3109/0954898X.2014.884126.

Alan Kan and Ruth Y. Litovsky. Binaural hearing with electrical stimulation. *Hearing Research*, 322:127–137, 2015. doi: 10.1016/j.heares.2014.08.005.

Adrien M. Khan, Ophir Handzel, Barbara J. Burgess, Diane Damian, Donald K. Eddington, and Joseph B. Nadol, Jr. Is word recognition correlated with the number of surviving spiral ganglion cells and electrode insertion depth in human subjects with cochlear implants? *The Laryngoscope*, 115(4):672–677, 2005. doi: 10.1097/01.mlg.0000161335.62139.80.

David C. Knill and Alexandre Pouget. The bayesian brain: the role of uncertainty in neural coding and computation. *Trends in Neurosciences*, 27(12):712–719, 2004. doi: 10.1016/j.tins.2004.10.007.

Anja Kurz, Fabian-Tobias Müller-Graff, Rudolf Hagen, and Kristen Rak. One click is not enough: Anatomy-based fitting in experienced cochlear implant users. *Otology & Neurotology*, 43(10):1176–1180, 2022. doi: 10.1097/MAO.0000000000003702.

Anja Kurz, Daniel Herrmann, Rudolf Hagen, and Kristen Rak. Using anatomy-based fitting to reduce frequency-to-place mismatch in experienced bilateral cochlear implant users: A promising concept. *Journal of Personalized Medicine*, 13(7):1109, 2023. doi: 10.3390/jpm13071109.

Imre Lahdelma and Tuomas Eerola. Register impacts perceptual consonance through roughness and sharpness. *Psychonomic Bulletin & Review*, 29:1700–1710, 2022. doi: 10.3758/s13423-021-02033-5.

Cynthia Lam, Brian C. J. Moore, Marina Salorio-Corbetto, and Deborah A. Vickers. The relationship between hearing experiences, music-listening behaviors, and chord-discrimination abilities for cochlear implant users. *Trends in Hearing*, 26:1–18, 2022. doi: 10.1177/23312165221143902.

Lara J. G. Lambriks, Marije van Hoof, Jeroen J. Briaire, Johan H. M. Frijns, Laura Hollink, Jeroen van Tongeren, Robert J. Stokroos, and Huib J. Theunisse. Individualized frequency fitting based on cochlear anatomy in experienced cochlear implant users. *Frontiers in Neurology*, 14:1168010, 2023a. doi: 10.3389/fneur.2023.1168010.

Lars Lambriks, Marc van Hoof, Joke Debruyne, Miranda Janssen, Janny Hof, Katja Hellingman, Elke Devocht, and Erwin George. Toward neural health measurements for cochlear implantation: The relationship among electrode positioning, the electrically evoked action potential, impedances and behavioral stimulation levels. *Frontiers in Neurology*, 14:1093265, 2023b. doi: 10.3389/fneur.2023.1093265.

Lars J. G. Lambriks, Marc van Hoof, Joke Debruyne, Miranda Janssen, Josef Chalupper, Katja van der Heijden, Jan R. Hof, Katja Hellingman, Elke Devocht, and Erwin George. Imaging-based frequency mapping for cochlear implants – evaluated using a daily randomized controlled trial. *Frontiers in Neuroscience*, 17:1119933, 2023c. doi: 10.3389/fnins.2023.1119933.

Preprint, not peer reviewed

David M. Landsberger, Maja Svrakic, J. Thomas Roland, and Mario Svirsky. The relationship between insertion angles, default frequency allocations, and spiral ganglion place pitch in cochlear implants. *Ear and Hearing*, 36(5):e207–e213, 2015. doi: 10.1097/AUD.0000000000000163.

Luis Lassaletta, Ana Castro, Maria Bastarrica, Rosa Pérez-Mora, Belen Herrán, Laura Sanz, Maria Jesús De Sarriá, and Javier Gavilán. Does music perception have an impact on quality of life following cochlear implantation? *Acta Oto-Laryngologica*, 128 (6):682–686, 2008. doi: 10.1080/00016480701675212.

Luis Lassaletta, Miryam Calvino, Isabel Sánchez-Cuadrado, and Javier Gavilán. Does it make any sense to fit cochlear implants according to the anatomy-based fitting? Our experience with the first series of patients. *Frontiers in Audiology and Otology*, 1: 1298538, 2023. doi: 10.3389/fauot.2023.1298538.

Diane S. Lazard, Christophe Vincent, Frédéric Venail, Paul Van de Heyning, Eric Truy, Olivier Sterkers, Piotr H. Skarzynski, Henryk Skarzynski, Karen Schauwers, Stephen O'Leary, et al. Pre-, per- and postoperative factors affecting performance of postlinguistically deaf adults using cochlear implants: A new conceptual model over time. *PLOS ONE*, 7(11):e48739, 2012.

E. Lehnhardt. Intracochlear placement of cochlear implant electrodes in soft surgery technique. *HNO*, 41(7):356–359, 1993.

Arne Leijon, Martin Dahlquist, and Karolina Smeds. Bayesian analysis of paired-comparison sound quality ratings. *The Journal of the Acoustical Society of America*, 146(5):3174–3183, 2019. doi: 10.1121/1.5131024.

Rebecca M Lewis, Melanie L Gilbert, Jordan A Beim, Andrew J Oxenham, and Charles J Limb. Using musical pitch interval comparisons to assess cochlear implant frequency-to-place maps. *Frontiers in Audiology and Otology*, 3:1565883, 2025. doi: 10.3389/fauot.2025.1565883.

Michael J. Lichtenstein, Fred H. Bess, and Shirley A. Logan. Diagnostic performance of the hearing handicap inventory for the elderly (screening version) against differing definitions of hearing loss. *Ear & Hearing*, 9(4):208–211, 1988. doi: 10.1097/00003446-198808000-00006.

Jason Tzu-Hsien Lien, Ben Williges, and Deborah Vickers. Cochlear implant remapping informed by measures of viability of the electrode-neural interface: a systematic review with meta-analysis. *Scientific Reports*, 15:09610, 2025. doi: 10.1038/s41598-025-09610-x.

Charles J. Limb and Jay T. Rubinstein. Current research on music perception in cochlear implant users. *Otolaryngologic Clinics of North America*, 45(1):129–140, 2012. doi: 10.1016/j.otc.2011.08.021.

Ruth Y. Litovsky. Spatial release from masking. *Acoustics Today*, 8(2):18–25, 2011.

Ruth Y. Litovsky, Peggy M. Johnstone, and Shelly P. Godar. Benefits of bilateral cochlear implants and/or hearing aids in children. *International Journal of Audiology*, 45(sup1): 78–91, 2006.

Preprint – not peer-reviewed

Ruth Y. Litovsky, Aaron Parkinson, and Jennifer Arcaroli. Spatial hearing and speech intelligibility in bilateral cochlear implant users. *Ear & Hearing*, 30(4):419–431, 2009. doi: 10.1097/AUD.0b013e3181a165be.

Gerald E. Loeb, Carolyn L. Byers, Stephen J. Rebscher, David E. Casey, Michael M. Fong, Robert A. Schindler, Robert F. Gray, and Michael M. Merzenich. Design and fabrication of an experimental cochlear prosthesis. *Medical and Biological Engineering and Computing*, 21(3):241–254, 1983. doi: 10.1007/BF02478489.

Christopher J. Long, Timothy A. Holden, Gary H. McClelland, William S. Parkinson, Clough Shelton, David C. Kelsall, and Zachary M. Smith. Examining the electro-neural interface of cochlear implant users using psychophysics, CT scans, and speech understanding. *Journal of the Association for Research in Otolaryngology*, 15(2):293–304, 2014. doi: 10.1007/s10162-013-0437-5.

Clarinda A. Makary, Jennifer Shin, Sharon G. Kujawa, M. Charles Liberman, and Saumil Bhatt. Age-related primary cochlear neuronal degeneration in human temporal bones. *Journal of the Association for Research in Otolaryngology*, 12(6):711–717, 2011. doi: 10.1007/s10162-011-0283-2.

Rajeev Mathew, Jaime Undurraga, Jingdan Shao, Robert P. Carlyon, Jessica Goodman, and Deborah Vickers. Electrode discrimination and speech perception in newly implanted cochlear implant users. *Hearing Research*, 354:151–163, 2017. doi: 10.1016/j.heares.2017.09.005.

Rajeev Mathew, Deborah Vickers, Patrick Boyle, Azhar Shaida, David Selvadurai, Dan Jiang, and Jaime Undurraga. Development of electrophysiological and behavioural measures of electrode discrimination in adult cochlear implant users. *Hearing Research*, 2018. doi: 10.1016/j.heares.2018.07.002.

Hugh J. McDermott. Music perception with cochlear implants: A review. *Trends in Amplification*, 8(2):49–82, 2004. doi: 10.1177/108471380400800203.

Theodore R. McRackan, Michael Bauschard, Jonathan L. Hatch, Emily Franko-Tobin, H. Ruth Droghini, Shaun A. Nguyen, and Judy R. Dubno. Meta-analysis of quality-of-life improvement after cochlear implantation and associations with speech recognition abilities. *Laryngoscope*, 128(4):982–990, 2018.

Catarina Mendonça and Symeon Delikaris-Manias. Statistical tests with ordinal data in spatial audio research. 2018. Paper 9987.

Jia Meng, Sha Li, Fei Zhang, Qian Li, and Zhao Qin. Cochlear size and shape variability and implications in cochlear implantation surgery. *Otology & Neurotology*, 37(9):1307–1313, 2016. doi: 10.1097/MAO.0000000000001189.

Griet Mertens, Ellen Andries, Andrea Kleine Punte, and Paul Van de Heyning. The smaller the frequency-to-place mismatch the better the hearing outcomes in cochlear implant recipients? *European Archives of Oto-Rhino-Laryngology*, 279(11):5243–5249, 2022. doi: 10.1007/s00405-021-07182-y.

Preprint - not peer-reviewed

Michael M. Merzenich, Robin P. Michelson, C. Robert Pettit, Robert A. Schindler, and Marion Reid. Neural encoding of sound sensation evoked by electrical stimulation of the acoustic nerve. In *Annals of Otology, Rhinology & Laryngology*, volume 82, pages 486–503, 1973. doi: 10.1177/000348947308200407.

Jonathan Mo, Nicole T. Jiam, Mickael L. D. Deroche, Patpong Jiradejvong, and Charles J. Limb. Effect of frequency response manipulations on musical sound quality for cochlear implant users. *Trends in Hearing*, 26:23312165221120017, 2022. doi: 10.1177/23312165221120017.

Aaron C. Moberly, Jonathan H. Lowenstein, Elizabeth Tarr, Amanda Caldwell-Tarr, D. Bradley Welling, Antoine J. Shahin, and Susan Nittrouer. Do adults with cochlear implants rely on different acoustic cues for phoneme perception than adults with normal hearing? *Journal of Speech, Language, and Hearing Research*, 57(2):566–582, 2014. doi: 10.1044/2013_JSLHR-H-13-0013.

Aaron C. Moberly, Jessica H. Lowenstein, and Susan Nittrouer. The enigma of poor performance by adults with cochlear implants. *Otology & Neurotology*, 37(10):1522–1528, 2016.

Brian C. J. Moore. Dead regions in the cochlea: conceptual foundations, diagnosis, and clinical applications. *Ear and Hearing*, 25(2):98–116, 2004. doi: 10.1097/01.AUD.0000120359.49711.D7.

Brian C. J. Moore and Chin-Tuan Tan. Perceived naturalness of spectrally distorted speech and music. *The Journal of the Acoustical Society of America*, 114(1):408–419, 2003. doi: 10.1121/1.1577552.

M. Moran, A. Rousset, and V. Looi. Music appreciation and music listening in prelingual and postlingually deaf adult cochlear implant recipients. *International Journal of Audiology*, 55(Suppl 2):S57–S63, 2016. doi: 10.3109/14992027.2016.1157630.

Isabelle Mosnier, Jean-Pierre Bebear, Mathieu Marx, Bernard Fraysse, Eric Truy, Geneviève Lina-Granade, Michel Mondain, Françoise Sterkers-Artières, Philippe Bordure, Alain Robier, Benoît Godey, Bernard Meyer, Bruno Frachet, Catherine Poncet-Wallet, Didier Bouccara, and Olivier Sterkers. Improvement of cognitive function after cochlear implantation in elderly patients. *JAMA Otolaryngology–Head & Neck Surgery*, 141(5):442–450, 2015. doi: 10.1001/jamaoto.2015.129.

Joseph B. Nadol. Patterns of neural degeneration in the human cochlea and auditory nerve: Implications for cochlear implantation. *Otolaryngology–Head and Neck Surgery*, 117(3):220–228, 1997. doi: 10.1016/S0194-5998(97)70178-5.

Leanne Nagels, Etienne Gaudrain, Deborah Vickers, Petra Hendriks, and Deniz Başkent. Prelingually deaf children with cochlear implants show better perception of voice cues and speech in competing speech than postlingually deaf adults with cochlear implants. *Ear & Hearing*, 45(4):952–968, 2024. doi: 10.1097/AUD.0000000000001489.

National Institute on Deafness and Other Communication Disorders. Cochlear implants, 2024. URL `https://www.nidcd.nih.gov/health/cochlear-implants`. Accessed: 2024.

Preprint – not peer-reviewed

John K. Niparko, Emily A. Tobey, Donna J. Thal, Laurie S. Eisenberg, Nae-Yuh Wang, Alexandra L. Quittner, and Nancy E. Fink. Spoken language development in children following cochlear implantation. *JAMA*, 303(15):1498–1506, 2010. doi: 10.1001/jama.2010.451.

Jack H. Noble, Robert F. Labadie, Renée H. Gifford, and Benoit M. Dawant. Image-guidance enables new methods for customizing cochlear implant stimulation strategies. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 21(5):820–829, 2013. doi: 10.1109/TNSRE.2013.2253333.

Jack H. Noble, René H. Gifford, Andrea J. Hedley-Williams, Benoit M. Dawant, and Robert F. Labadie. Clinical evaluation of an image-guided cochlear implant programming strategy. *Audiology and Neurotology*, 19(3):152–160, 2020.

Sylvie Nozaradan, Isabelle Peretz, Peter E. Keller, Peter Vuust, and André Mouraux. Neural tracking of the musical beat is enhanced by low-frequency sounds. *Proceedings of the National Academy of Sciences*, 115(32):8221–8226, 2018. doi: 10.1073/pnas.1801421115. CHECK: verify full author list.

Heidi Olze, Agnieszka J. Szczepek, Heidemarie Haupt, Ulrike Förster, Nora Zirke, Silke Gräbel, and Birgit Mazurek. Cochlear implantation has a positive influence on quality of life, tinnitus, and psychological comorbidity. *The Laryngoscope*, 121(10):2220–2227, 2011. doi: 10.1002/lary.22145.

Richard Parncutt. Revision of Terhardt's psychoacoustical model of the root(s) of a musical chord. *Music Perception*, 6(1):65–93, 1988. doi: 10.2307/40285416. CHECK doi.

Gordon E Peterson and Harold L Barney. Control methods used in a study of the vowels. *The Journal of the Acoustical Society of America*, 24(2):175–184, 1952. doi: 10.1121/1.1906875.

Nathaniel R. Peterson, David B. Pisoni, and Richard T. Miyamoto. Cochlear implants and spoken language processing abilities: review and assessment of the literature. *Restorative Neurology and Neuroscience*, 28(2):237–250, 2010. doi: 10.3233/RNN-2010-0535.

Bryan E. Pfingst, Deborah J. Colesa, Sara Hembrador, Sung Yun Kang, John C. Middlebrooks, Yehoash Raphael, and Grace L. Su. Detection of pulse trains in the electrically stimulated cochlea: effects of cochlear health. *Journal of the Acoustical Society of America*, 130(6):3954–3968, 2011a. doi: 10.1121/1.3651820.

Bryan E. Pfingst, Deborah J. Colesa, Stefan Hembrador, Su-Wooi Kang, John C. Middlebrooks, Yehoash Raphael, and Gina L. Su. Detection of pulse trains in the electrically stimulated cochlea: Effects of cochlear health. *The Journal of the Acoustical Society of America*, 130(6):3954–3968, 2011b. doi: 10.1121/1.3651820.

Bryan E Pfingst, Ning Zhou, Deborah J Colesa, Margaret M Watts, Stefan B Strahl, Yehoash Raphael, Richard A Altschuler, and Suyash Bhatt. Importance of cochlear health for implant function. *Hearing Research*, 322:77–88, 2015. doi: 10.1016/j.heares.2014.09.009.

Preprint - not peer-reviewed

Reinier Plomp and Willem J. M. Levelt. Tonal consonance and critical bandwidth. *Journal of the Acoustical Society of America*, 38(4):548–560, 1965. doi: 10.1121/1.1909741.

Frank Rattay, Patrick Lutter, and Hubert Felix. A model of the electrically excited human cochlear neuron. I. Contribution of neural substructures to the generation and propagation of spikes. *Hearing Research*, 153(1–2):43–63, 2001. doi: 10.1016/S0378-5955(00)00256-2.

Lina A. J. Reiss, Christopher W. Turner, Sandy R. Erenberg, and Bruce J. Gantz. Changes in pitch with a cochlear implant over time. *Journal of the Association for Research in Otolaryngology*, 8(2):241–257, 2007. doi: 10.1007/s10162-007-0077-8.

Lina A. J. Reiss, Courtney S. Shayman, Emily P. Walker, Keri O. Bennett, Jennifer R. Fowler, Christy L. Hartling, Beth Glickman, Michael R. Lasarev, and Yonghee Oh. Binaural pitch fusion: Comparison of normal-hearing and hearing-impaired listeners. *The Journal of the Acoustical Society of America*, 143(3):1642–1653, 2018. doi: 10.1121/1.5026934.

Sara K Röbler, Blake P O'Connell, Jacob B Hunter, René H Gifford, Colin L W Driscoll, and Matthew L Carlson. Electrocochleography and cognition are important predictors of speech perception outcomes in noise for cochlear implant recipients. *Scientific Reports*, 12:3083, 2022. doi: 10.1038/s41598-022-07175-7.

Alexandre T. Roy, Martin J. Penner, Anuj Bhatt, Vijayalakshmi Rao, Chan-Chan Lau, and Fan-Gang Zeng. Sound quality after cochlear implantation: effects of electrode array position. *Ear & Hearing*, 37(2):e100–e109, 2016. doi: 10.1097/AUD.0000000000000244.

Alexis T. Roy, Patpong Jiradejvong, Courtney Carver, and Charles J. Limb. Assessment of sound quality perception in cochlear implant users during music listening. *Otology & Neurotology*, 33(3):319–327, 2012. doi: 10.1097/MAO.0b013e31824296a9.

Eitan Sagi and Mario A. Svirsky. A level-adjusted cochlear frequency-to-place map for estimating tonotopic frequency mismatch with a cochlear implant. *Ear and Hearing*, 46(4):963–975, 2025. doi: 10.1097/AUD.0000000000001641.

Reinhold Schatzer, Katrien Vermeire, Dimphna Visser, Andreas Krenmayr, Michael Kals, Marc Voormolen, Paul Van de Heyning, and Clemens Zierhofer. Electric-acoustic pitch comparisons in single-sided-deaf cochlear implant users: Frequency-place functions and rate pitch. *Hearing Research*, 309:26–35, 2014. doi: 10.1016/j.heares.2013.11.003.

Kara C. Schvartz-Leyzac and Bryan E. Pfingst. Across-site patterns of electrically evoked compound action potential amplitude-growth functions in multichannel cochlear implant recipients and the effects of the interphase gap. *Hearing Research*, 341:50–65, 2016. doi: 10.1016/j.heares.2016.08.002.

Robert V. Shannon, Fan-Gang Zeng, Vinay Kamath, John Wygonski, and Michael Ekelid. Speech recognition with primarily temporal cues. *Science*, 270(5234):303–304, 1995. doi: 10.1126/science.270.5234.303.

Preprint - not peer-reviewed

Margaret W. Skinner, Timothy A. Holden, Barbara R. Whiting, Arne H. Voie, Barry Brunsden, John G. Neely, Eric A. Saxon, Timothy E. Hullar, and Robert A. Petrel. In vivo estimates of the position of Advanced Bionics electrode arrays in the human cochlea. *Annals of Otology, Rhinology & Laryngology*, 111(9 Suppl):2–24, 2002. doi: 10.1177/00034894021110S901.

Anthony J. Spahr, Michael F. Dorman, Leonid M. Litvak, Sasha Van Wie, Renée H. Gifford, Philipos C. Loizou, Louise M. Loiselle, Terri Oakes, and Sara Cook. Development and validation of the AzBio sentence lists. *Ear and Hearing*, 33(1):112–117, 2012. doi: 10.1097/AUD.0b013e31822c2549.

Aniket G. Srinivasan, Mario Padilla, Robert V. Shannon, and David M. Landsberger. Improving speech perception in noise with current focusing in cochlear implant users. *Hearing Research*, 299:29–36, 2013. doi: 10.1016/j.heares.2013.02.004.

Olga Stakhovskaya, Dhananjay Sridhar, Benson H. Bonham, and Patricia A. Leake. Frequency map for the human cochlear spiral ganglion: implications for cochlear implants. *Journal of the Association for Research in Otolaryngology*, 8(2):220–233, 2007a.

Olga Stakhovskaya, Dhiresha Sridhar, Ben H Bonham, and Patricia A Leake. Frequency map for the human cochlear spiral ganglion: implications for cochlear implants. *Journal of the Association for Research in Otolaryngology*, 8(2):220–233, 2007b. doi: 10.1007/s10162-007-0076-9.

Jacob J. Sturm, Viraj P. Patel, Graham Dibelius, Manfred Kuhlmey, and Ana H. Kim. Frequency-to-place mismatch impacts cochlear implant quality of life, but not speech recognition. *The Laryngoscope*, 134(6):2820–2827, 2024. doi: 10.1002/lary.31264.

Anita Sulibhavi, Christopher Z. Wen, Kevin L. Li, and Tiffany P. Hwa. The impact of cochlear implant experience on music perception: Systematic review and meta-analysis. *Current Otorhinolaryngology Reports*, 10:315–321, 2022. doi: 10.1007/s40136-022-00418-1.

Mario A. Svirsky, Adriana Silveira, Héctor Suarez, Harold Neuburger, Thanh T. Lai, J. B. Simmons, and Anne L. Beiter. Auditory learning and adaptation after cochlear implantation: a preliminary study of discrimination and labelling of vowel sounds by cochlear implant users. *Acta Oto-Laryngologica*, 125(2):144–153, 2005.

Mario A Svirsky, Ana Silveira, Howard Neuburger, Howard Suarez, and Tina T Lai. A model of incomplete adaptation to a severely shifted frequency-to-electrode mapping by cochlear implant users. *Journal of the Association for Research in Otolaryngology*, 11(4):565–579, 2010. doi: 10.1007/s10162-010-0214-x.

Ernst Terhardt. The concept of musical consonance: a link between music and psychoacoustics. *Music Perception*, 1(3):276–295, 1984. doi: 10.2307/40285261. CHECK doi.

The MathWorks Inc. MATLAB r2023a, 2023. URL `https://www.mathworks.com`.

Thilo Thiede, William C. Treurniet, Roland Bitto, Christian Schmidmer, Thomas Sporer, Jannes G. Beerens, Catherine Conseldo, Michael Keyhl, Gerhard Stoll, Karlheinz Brandenburg, and Bernd Feiten. PEAQ – the ITU standard for objective measurement of perceived audio quality. *Journal of the Audio Engineering Society*, 48(1-2):3–29, 2000.

Preprint – not peer-reviewed

Chandra S. Throckmorton and Leslie M. Collins. Investigation of the effects of temporal and spatial interactions on speech-recognition skills in cochlear-implant subjects. *The Journal of the Acoustical Society of America*, 105(2):861–873, 1999. doi: 10.1121/1.426275.

Y. C. Tong, R. C. Black, G. M. Clark, I. C. Forster, R. C. Dowell, L. F. A. Martin, and J. F. Patrick. A preliminary report on a multiple-channel cochlear implant operation. *The Journal of Laryngology and Otology*, 93(7):679–695, 1979. doi: 10.1017/S0022215100087771.

Yit Chow Tong, Graeme M. Clark, Peter J. Blamey, Peter A. Busby, and Richard C. Dowell. Psychophysical studies for two multiple-channel cochlear implant patients. *The Journal of the Acoustical Society of America*, 71(1):153–160, 1982. doi: 10.1121/1.387342.

Frederic Venail, Caroline Mathiolon, Sophie Menjot de Champfleur, Jean Pierre Piron, Marielle Sicard, Françoise Villemus, Marie Aude Vessigaud, Françoise Sterkers-Artieres, Michel Mondain, and Alain Uziel. Effects of Electrode Array Length on Frequency-Place Mismatch and Speech Perception with Cochlear Implants. *Audiology and Neurotology*, 20(2):102–111, 2015a. ISSN 1420-3030, 1421-9700. doi: 10.1159/000369333. URL https://www.karger.com/Article/FullText/369333.

Frédéric Venail, Camille Mathiolon, Sophie Menjot de Champfleur, Jean-Pierre Piron, Stéphanie Siber, Françoise Artieres-Roux, Alain Uziel, and Michel Mondain. Effects of electrode array length on frequency-place mismatch and speech perception with cochlear implants. *Audiology and Neurotology*, 20(3):171–182, 2015b. doi: 10.1159/000369333.

Alessandro Volta. On the electricity excited by the mere contact of conducting substances of different kinds. *Philosophical Transactions of the Royal Society of London*, 90:403–431, 1800.

C. von Ilberg, J. Kiefer, J. Tillein, T. Pfenningdorff, R. Hartmann, E. Stürzebecher, and R. Klinke. Electric-acoustic stimulation of the auditory system: new technology for severe hearing loss. *ORL: Journal for Oto-Rhino-Laryngology and Its Related Specialties*, 61(6):334–340, 1999. doi: 10.1159/000027695.

Amit Walia, Amanda J. Ortmann, Sara Lefler, Timothy A. Holden, Sidharth V. Puram, Jacques A. Herzog, and Craig A. Buchman. Electrocochleography-based tonotopic map: II. Frequency-to-place mismatch impacts speech-perception outcomes in cochlear implant recipients. *Ear and Hearing*, 46(1):265–276, 2025. doi: 10.1097/AUD.0000000000001579.

Dorji Wangchuk, Yoon Ah Kim, Sung Hoon Choi, and Il Joon Moon. Anatomy-based fitting in cochlear implant users: Effect on speech perception and subjective outcomes. *Clinical and Experimental Otorhinolaryngology*, 17(2):161–170, 2024. doi: 10.21053/ceo.2023.01234.

Ben Williges and Tim Jürgens. A flexible vocoder toolbox for acoustic simulations of cochlear implants in MATLAB. *arXiv preprint*, 2019. arXiv:1910.02541.

Preprint - not peer-reviewed

Blake S. Wilson and Michael F. Dorman. Cochlear implants: a remarkable past and a brilliant future. *Hearing Research*, 242(1-2):3–21, 2007. doi: 10.1016/j.heares.2007.11.001.

Blake S. Wilson and Michael F. Dorman. Cochlear implants: A remarkable past and a brilliant future. *Hearing Research*, 242(1–2):3–21, 2008. doi: 10.1016/j.heares.2008.06.005.

Blake S. Wilson, Charles C. Finley, Dewey T. Lawson, Robert D. Wolford, Donald K. Eddington, and William M. Rabinowitz. Better speech recognition with cochlear implants. *Nature*, 352:236–238, 1991.

Ruijie Yang, Ting Xu, Yang Liu, Chun Chen, Zhigang Wang, and Yanmei Li. The effect of anatomically based frequency allocation on speech perception of cochlear implant users: A systematic review and meta-analysis. *Frontiers in Neuroscience*, 16:834036, 2022. doi: 10.3389/fnins.2022.834036.

Diego Zanetti, Maurizio Guida, Giulia Bianchin, Andrea Frisina, and Marco Guglieri. Tonotopic fitting in cochlear implant users: a prospective, double-blind, randomised crossover study. *Acta Oto-Laryngologica*, 141(5):449–455, 2021. doi: 10.1080/00016489.2021.1875462.

Robert J. Zatorre and Valorie N. Salimpoor. From perception to pleasure: Music and its neural substrates. *Proceedings of the National Academy of Sciences*, 110(Suppl. 2):10430–10437, 2013. doi: 10.1073/pnas.1301228110.

Fan-Gang Zeng, Stephen Rebscher, William Harrison, Xiaoan Sun, and Haihong Feng. Cochlear implants: System design, integration, and evaluation. *IEEE Reviews in Biomedical Engineering*, 1:115–142, 2008. doi: 10.1109/RBME.2008.2008250.

Sławomir K. Zieliński, Francis Rumsey, and Søren Bech. On some biases encountered in modern audio quality listening tests — a review. *Journal of the Audio Engineering Society*, 56(6):427–451, 2007.

Vera Zimmer, Jürgen L. Verhey, Michael Ziese, and Maike Böckmann-Barthel. Harmony perception in prelingually deaf, juvenile cochlear implant users. *Frontiers in Neuroscience*, 13:466, 2019. doi: 10.3389/fnins.2019.00466.

Preprint - not peer-reviewed